Globalization with Capital Integration and Spatial Entry *

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Abstract

Can Tariffs Be Good? I extend the Melitz (2003)–Chaney (2008)–Caliendo et al. (2023) framework by introducing an integrated capital market and show that a unilateral tariff increase can induce capital inflows, which in turn promote firm entry and potentially generate welfare gains for the tariff-imposing country. I quantify these mechanisms by calibrating the model to U.S. state-level and Chinese trade data and applying it to the 2018 U.S. tariffs on Chinese imports. Model-predicted firm entry is positively correlated with subsequent establishment growth across U.S. states. The counterfactual analysis predicts welfare gains for workers in all U.S. states and for households receiving both labor and capital income in most states. Interestingly, given the prominence of tariffs as a campaign promise in the 2016 presidential election, states with larger predicted welfare gains experienced greater increases in Republican vote share between the 2012 and 2016 presidential elections.

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1 Introduction

Economists have long warned against tariffs. Politicians have continued to champion them. And now, as protectionism makes a comeback around the world, an old debate is taking center stage once again: Can tariffs actually be good? In another attempt to answer this question, I extend the Melitz (2003)–Chaney (2008)–Caliendo et al. (2023) framework to incorporate capital mobility across regions and show that a unilateral tariff increase can induce capital inflows, which in turn promote firm entry and potentially generate welfare gains even in the absence of tariff revenue.

A tariff induces factor movements through changes in relative factor returns. A tariff increase shifts Home expenditure toward Home firms, raising the returns to both factors (labor and capital) in Home relative to Foreign. The resulting higher return to capital in Home induces capital to flow from Foreign to Home. The increase in Home factor returns is closely related to the terms-of-trade mechanism underlying the classic optimal-tariff argument (Johnson, 1953; Bagwell and Staiger, 1999). By reducing its demand for Foreign goods relative to their supply and its supply of Home goods to the world market relative to their demand, Home induces an increase in the relative price of its goods. This terms-of-trade improvement translates into higher returns to both labor and capital in Home relative to Foreign. This mechanism differs from Mundell (1957), where, in a Heckscher–Ohlin framework, factor returns follow the Stolper–Samuelson mechanism and thus respond to changes in the relative prices of goods with different factor intensities.

The resulting capital inflow, in turn, promotes firm entry through a straightforward mechanism. From the perspective of a potential entrant, entry depends on the value of the market relative to the cost of establishing a firm. With homothetic production technologies, market value scales proportionally with wages, whereas fixed entry costs rise less than proportionally with wages as long as they are not entirely labor-based. By raising the wage–rental ratio, capital inflows therefore increase the value of entry relative to its cost, encouraging additional firm entry. The increase in the number of Home firms under trade protection is consistent

with Venables (1985), Bagwell and Staiger (2012), and Melitz and Ottaviano (2008), but the underlying mechanism differs. In these studies, trade protection induces firms to relocate toward Home through changes in markups in the absence of capital mobility. In contrast, markups are constant in my framework, and the increase in the number of Home firms is driven by new entry induced by capital inflows.

This entry-margin response plays a central role in the model's welfare implications. In its absence, capital inflows raise wages relative to rental rates as capital becomes more abundant, but the resulting positive income effect is insufficient to offset the tariff-induced increase in consumer prices. The reason is that the tariff raises the price index directly, whereas the positive income effect operates only indirectly through the equilibrium adjustment of factor returns.[1] With endogenous entry, by contrast, the emergence of new domestic varieties dampens the tariff-induced increase in the price index.[2] Combined with the positive income effect from capital inflows, this dampened price response can generate welfare gains for domestic agents following a unilateral tariff increase, even in the absence of tariff revenue. This result contrasts with the standard theoretical prediction that tariffs reduce welfare in the absence of tariff revenue, as characterized by Arkolakis, Costinot, and Rodríguez-Clare (2012).

The interaction among trade policy, firm location, and international capital movements has long been studied, primarily through the lens of how trade policy shapes firms' location decisions and how the resulting relocation of firms drives international factor movements. This line of inquiry follows the mechanism introduced by Krugman (1991), whereby changes in market access induce firms to relocate, with capital following their location decisions. Martin and Rogers (1995) and Boulhol (2009) assume a one-to-one mapping between firms

[1] To show that capital mobility alone is insufficient to offset the welfare loss from a tariff increase, I introduce capital mobility into the Armington framework (Appendix J), which lacks an entry margin, and show that a unilateral tariff increase leads to an unambiguous welfare loss despite inducing capital inflows.

[2] When capital intensity is sufficiently high, the entry response can be strong enough to reverse the tariff-induced increase in the price index.

and capital, whereas Brainard (1993), Horstmann and Markusen (1992), and Motta (1992) consider a more nuanced relationship in which firms choose between exporting and foreign direct investment (FDI). The latter models formalize the tariff-jumping FDI mechanism studied in the empirical literature. Because these approaches emphasize firms' location and organizational decisions, trade policy can alter the equilibrium market structure, making it difficult to derive general welfare predictions. This paper instead explores a complementary mechanism in which trade policy induces capital movements, which in turn encourage or discourage firm entry and shape welfare outcomes. By abstracting from firms' strategic location decisions, the framework isolates the role of capital mobility and endogenous entry, allowing their welfare implications to be characterized more transparently.

Regarding the specific model setup, this paper builds on the Melitz (2003)–Chaney (2008) framework with roundabout production developed by Caliendo et al. (2023), and introduces an integrated capital market in which capital flows frictionlessly across locations while labor remains geographically immobile. Tariff revenue is assumed to be dissipated. Although capital is mobile across locations for production purposes, its ownership is fixed and exogenous. Household welfare is therefore based on factor ownership rather than the location of production: rental income generated by foreign-owned capital employed in Home accrues to Foreign households and is not included in Home household income.

Section 2 illustrates the key mechanism in a symmetric two-country setting through two limiting cases: one with no capital as a production input and one with no intermediate inputs. With roundabout production alone, the entry response is proportional to the change in the real wage and thus amplifies the standard welfare effect of trade policy. In contrast, when capital and labor are the only factors of production, the entry response is proportional to the change in the nominal wage relative to the rental rate. Because this ratio increases with the tariff, firm entry dampens the standard price-increasing effect of a tariff increase. The general-case entry response is a convex combination of the two limiting cases, with the welfare outcome similarly interpolating between them.

Section 3 calibrates the multi-region version of the model and employs exact hat algebra to analyze President Trump's 2018 unilateral tariffs on Chinese imports. The calibration uses a region-by-region trade matrix constructed from the Freight Analysis Framework (FAF) and USA Trade Online data. The counterfactual results, reported in Section 3.3, show that the model-predicted changes in the number of firms are positively correlated with subsequent changes in the number of establishments reported in the County Business Patterns across U.S. states. In terms of welfare, households in most U.S. states experience gains when they are assumed to receive both labor and capital income. Section 4 further shows that the cross-state distribution of predicted welfare changes is positively correlated with changes in Republican vote shares between the 2012 and 2016 U.S. presidential elections.

The counterfactual welfare results contrast with those of Fajgelbaum et al. (2020) and Caliendo and Parro (2021), who find aggregate welfare losses from the 2018 tariffs. At the same time, the prediction that workers and households can gain from tariffs resonates with the broader literature on the adverse effects of import competition, beginning with Autor, Dorn, and Hanson (2013). While Caliendo, Dvorkin, and Parro (2019) emphasizes labor-market adjustment frictions and Galle, Rodríguez-Clare, and Yi (2023) shows how worker heterogeneity can generate welfare losses from trade despite aggregate gains, this paper provides an alternative theoretical mechanism through which trade protection can improve welfare for both workers and households.

2 Two-Country Model

To illustrate the key mechanisms and their underlying intuition, I begin with a two-country, one-sector model. In Section 3, I extend the model to multiple regions and calibrate it to the data.

2.1 Model setup

There are two countries, H and F, and a single sector. Firms share a common Cobb–Douglas production technology but differ in productivity z. A firm with productivity z produces

$$y = z \left(\frac{L}{\xi}\right)^{\xi} \left[\left(\frac{E}{\phi(1-\xi)}\right)^{\phi} \left(\frac{K}{(1-\phi)(1-\xi)}\right)^{1-\phi}\right]^{1-\xi},$$

where L, K, and E denote labor, capital, and intermediate inputs, respectively. When $\phi = 1$, the model reduces to the Caliendo et al. (2023); when $\phi = 0$, it reduces to the Melitz (2003)-Chaney (2008) with just mobile capital.

Labor is immobile across countries, with endowments L_H and L_F, while capital is perfectly mobile. The world capital stock is K_0, of which households in H and F own fixed shares γ and $1-\gamma$, respectively. Capital can be employed in either country without friction, equalizing the rental rate r across locations. I choose r as the numeraire.

2.2 Households' demand

Households in both countries have identical CES preferences over the set of available varieties Ω_n,

$$U_n = Q_n^c = \left(\int_{\omega\in\Omega_n} q_n^c(\omega)^{\frac{\sigma-1}{\sigma}}\, d\omega\right)^{\frac{\sigma}{\sigma-1}}.$$

Household income is $I_n = W_n L_n + rK_n$, where $K_H = \gamma K_0$ and $K_F = (1-\gamma)K_0$. Utility maximization yields demand

$$q_n^c(\omega) = p_n(\omega)^{-\sigma} P_n^{\sigma-1} I_n,$$

where the corresponding price index is $P_n = \left(\int_{\omega\in\Omega_n} p_n(\omega)^{1-\sigma}\, d\omega\right)^{\frac{1}{1-\sigma}}$.

2.3 Producers' demand

Firms use the same CES aggregate as an intermediate input. Given intermediate-input expenditure $M_n = P_n E_n$, cost minimization yields demand for variety ω,

$$q_n^e(\omega) = p_n(\omega)^{-\sigma} P_n^{\sigma-1} M_n,$$

where P_n is the CES price index defined above.

2.4 Supply

Firms draw productivity z prior to production from a Pareto distribution,

$$\Pr(Z \geq z) = \left(\frac{\underline{z}}{z}\right)^{\alpha}, \qquad z \geq \underline{z}, \qquad \alpha > \sigma - 1,$$

which is identical across countries in the symmetric case. After observing z, firms decide whether to operate, export, or exit.

Trade from country s to country n is subject to an iceberg trade cost $1+\tau_{sn}$. Trade costs generate no tariff revenue. Since varieties are uniquely associated with productivity draws, I henceforth index prices and quantities by source s, destination n, and productivity z. Let

$$c_s \equiv W_s^{\xi}(P_s^{\phi} r^{1-\phi})^{1-\xi}$$

denote the unit cost of a firm with unit productivity in country s. Given total demand $q_{sn}(z) = q_{sn}^c(z) + q_{sn}^e(z)$, cost minimization implies

$$K_{sn}(z) = \frac{(1-\xi)(1-\phi)}{r}\frac{c_s}{z} q_{sn}(z)(1+\tau_{sn}),$$

$$E_{sn}(z) = \frac{(1-\xi)\phi}{P_s}\frac{c_s}{z} q_{sn}(z)(1+\tau_{sn}), \qquad L_{sn}(z) = \frac{\xi}{W_s}\frac{c_s}{z} q_{sn}(z)(1+\tau_{sn}).$$

Under monopolistic competition, firms charge a constant markup over marginal cost, so that

$$p_{sn}(z) = \frac{\sigma}{\sigma-1}\frac{c_s}{z}(1+\tau_{sn}).$$

2.5 Aggregate values

The value of trade for a variety produced by a firm with productivity z located in country s and sold in country n is thus

$$V_{sn}(z) = p_{sn}(z)\, q_{sn}(z) = \sigma B_n \left((1+\tau_{sn})\frac{c_s}{z}\right)^{1-\sigma}, \tag{1}$$

where

$$B_n = \frac{W_n L_n + r K_n}{P_n^{1-\sigma}} \left(\frac{1}{\sigma}\right)^{\sigma} \left(\frac{1}{\sigma - 1}\right)^{1-\sigma} \tag{2}$$

B_n is interpreted as a measure of the *market size*. It increases with aggregate income and with the price index. Higher income raises demand for each variety, while a higher price index reflects weaker competition across varieties, which also increases the demand faced by individual firms. As a result, a larger measure of market size corresponds to a more favorable market environment for firm survival.

2.5.1 Technology threshold for selling in a market

A firm located in country s serves market n only if variable profits cover the fixed cost of selling there:

$$\pi_{sn}(z) \geq F_{sn} c_s,$$

where F_{sn} denotes the fixed-cost shifter for serving market n. Using this expression, variable profits are

$$\pi_{sn}(z) = V_{sn}(z) - W_s L_{sn}(z) - P_s E_{sn}(z) - r K_{sn}(z) = \frac{V_{sn}(z)}{\sigma}.$$

The productivity cutoff for a firm from country s to serve market n is therefore

$$z_{\min}^{sn} = \left(\frac{F_{sn}}{B_n}\right)^{\frac{1}{\sigma-1}} c_s^{\frac{\sigma}{\sigma-1}} (1 + \tau_{sn}). \tag{3}$$

Post–fixed-cost profits from serving market n are

$$\tilde{\pi}_{sn}(z) = \pi_{sn}(z) - F_{sn} c_s.$$

2.5.2 Aggregate trade value

Let Ω_{sn} denote the set of firms in country s that serve market n. Aggregate trade value from s to n is

$$V_{sn} = N_s \Pr(\tilde{\pi}_{sn} > 0) \mathbf{E}[V_{sn}(z) \mid \tilde{\pi}_{sn} > 0],$$

where N_s denotes the mass of firms in country s. Using the Pareto productivity distribution yields

$$V_{sn} = \frac{\sigma\alpha \underline{z}^{\alpha}}{\alpha + 1 - \sigma} N_s B_n^{\frac{\alpha}{\sigma-1}} F_{sn}^{1+\frac{\alpha}{1-\sigma}} c_s^{1-\epsilon} (1+\tau_{sn})^{-\alpha}, \tag{4}$$

where $\epsilon \equiv \frac{\sigma\alpha}{\sigma-1}$.

Aggregate post–fixed-cost profits from serving market n are

$$\tilde{\pi}_{sn} = \int_{\omega\in\Omega_{sn}} \left[\pi_{sn}(\omega) - F_{sn}c_s\right] d\omega = \frac{\sigma-1}{\sigma\alpha} V_{sn}. \tag{5}$$

Equation (4) follows the standard Melitz–Chaney formulation, with the wage replaced by the composite unit cost c_s, reflecting the use of labor, capital, and intermediate inputs in production.

2.6 Equilibrium conditions

An equilibrium is a collection of wages $\{W_n\}$, price indices $\{P_n\}$, and firm masses $\{N_n\}$, for $n \in \{H, F\}$, such that households optimize, firms maximize profits, and labor, capital, and goods markets clear, taking the common rental rate r as the numéraire.

The six endogenous variables

$$\{W_H, W_F, N_H, N_F, P_H, P_F\}$$

are jointly determined by two free-entry conditions, two labor-market-clearing conditions, the global capital-market-clearing condition, and one trade-balance condition, with the other implied by Walras' law.

Free entry. For each country $s \in \{H, F\}$, free entry requires aggregate post–fixed-cost profits to equal total entry costs:

$$\sum_{n\in\{H,F\}} \tilde{\pi}_{sn} = \frac{\sigma-1}{\sigma\alpha} \sum_{n\in\{H,F\}} V_{sn} = N_s F_s^E c_s,$$

where F_s^E denotes the fixed-cost shifter for entry and $\tilde{\pi}_{sn}$ denotes aggregate post–fixed-cost profits from serving market n, as defined in Equation 5.

Labor market clearing. Labor in country s is used for variable production, fixed market-access costs, and entry. Let $\tilde{L}_{sn}$ denote labor used by firms from s to serve market n, including variable and fixed selling costs:

$$\tilde{L}_{sn} = \xi \frac{V_{sn}}{\sigma W_s} \left(\sigma - 1 + \frac{\alpha + 1 - \sigma}{\alpha} \right).$$

Labor-market clearing requires

$$L_s = \sum_n \tilde{L}_{sn} + N_s \frac{\xi F_s^E c_s}{W_s}.$$

Using the free-entry condition, this simplifies to

$$W_s L_s = \xi \sum_n V_{sn}, \qquad s \in \{H, F\}.$$

Capital market clearing. With frictionless capital mobility, capital is allocated globally across variable production, fixed market-access costs, and entry. Let $\tilde{K}_{sn}$ denote capital used by firms from s to serve market n, including variable and fixed selling costs:

$$\tilde{K}_{sn} = \frac{(1-\xi)(1-\phi)V_{sn}}{\sigma r} \left(\sigma - 1 + \frac{\alpha + 1 - \sigma}{\alpha} \right).$$

Global capital-market clearing requires

$$K_0 = \sum_{s,n} \tilde{K}_{sn} + \sum_s N_s \frac{(1-\xi)(1-\phi)F_s^E c_s}{r}.$$

Using the free-entry condition, this simplifies to

$$rK_0 = (1-\xi)(1-\phi) \sum_{s,n} V_{sn}.$$

Capital employed in country s generally differs from the capital owned by its households, $K_H = \gamma K_0$ and $K_F = (1-\gamma)K_0$, since capital is freely mobile across countries.

Intermediate inputs. Intermediate inputs account for a share $\phi(1-\xi)$ of firms' total expenditure. Hence,

$$M_s = P_s E_s = \phi(1-\xi) \sum_n V_{sn}, \qquad s \in \{H, F\}.$$

Combining this with the labor- and capital-market-clearing conditions yields

$$W_H L_H + W_F L_F + rK_0 = \sum_{s,n} V_{sn},$$

so that aggregate world income equals aggregate world expenditure.

Goods market clearing and trade balance. Trade-balance conditions require that total expenditure equals total income within each country. For country H,

$$W_H L_H + \gamma r K_0 + P_H E_H = V_{HH} + V_{FH},$$

and for country F,

$$W_F L_F + (1-\gamma) r K_0 + P_F E_F = V_{FF} + V_{HF}.$$

Number of firms. Combining free entry with labor-market clearing yields the equilibrium mass of firms:

$$N_s = \frac{L_s}{\xi F_s^E} \frac{\sigma - 1}{\sigma \alpha} W_s^{1-\xi} \left(P_s^{\phi} r^{1-\phi} \right)^{\xi - 1}. \tag{6}$$

Thus, N_s is determined by W_s and P_s. The free-entry conditions can therefore be substituted out, reducing the equilibrium system to the four endogenous variables $\{W_H, W_F, P_H, P_F\}$.

Summary. The equilibrium is characterized by the following four equations:

$$L_H = \frac{\xi}{W_H} \sum_n V_{Hn}, \tag{7}$$

$$L_F = \frac{\xi}{W_F} \sum_n V_{Fn}, \tag{8}$$

$$K_0 = \frac{(1-\xi)(1-\phi)}{r} \sum_n \left(V_{Hn} + V_{Fn} \right), \tag{9}$$

$$W_H L_H (1 + \phi \frac{1-\xi}{\xi}) + \gamma r K_0 = V_{HH} + V_{FH}. \tag{10}$$

2.7 Wage–Entry linkage

Before turning to the general analysis, I build intuition for the mechanism governing firm entry by considering two limiting cases of the model: $\phi = 1$ and $\phi = 0$.

2.7.1 Production with labor and intermediate inputs only: $\phi = 1$

When $\phi = 1$, capital is not used in production, and all production inputs are cleared domestically. This limiting case closely follows the setup in Caliendo et al. (2023). The equilibrium measure of firms simplifies to

$$N_s = \frac{L_s}{\xi F_s^E} \frac{\sigma - 1}{\sigma \alpha} \left(\frac{W_s}{P_s} \right)^{1-\xi}. \tag{11}$$

In the original Melitz-Chaney model, where $\xi = 1$, the equilibrium measure of firms is pinned down by the labor endowment and structural parameters. In contrast, with roundabout production, the measure of firms also depends on endogenous wages and the price index. In particular, the expression above shows that the equilibrium measure of firms increases with the real wage, $\frac{W_s}{P_s}$, which also determines welfare. Thus, the endogenous entry response arising from roundabout production amplifies the welfare effects present in the standard Melitz-Chaney model, as an increase in real wages induces additional firm entry and expands the set of available varieties.

2.7.2 Production with labor and mobile capital only: $\phi = 0$

In this case, there are no intermediate inputs, while capital can move freely across countries and the rental rate r is chosen as the numeraire. Under these assumptions, the equilibrium measure of firms is given by

$$N_s = \frac{L_s}{\xi F_s^E} \frac{\sigma - 1}{\sigma \alpha} r^{\xi - 1} W_s^{1-\xi}. \tag{12}$$

The measure of firms now responds to changes in the nominal wage, in contrast to the case with roundabout production, where it responds to changes in the real wage. More generally, Equation (12) shows that firm entry is increasing in the relative price of labor, W_s/r: as labor becomes more expensive relative to capital, the equilibrium measure of firms increases.

This result follows directly from the tradeoff between the cost of entry and the market value of a firm after entry. The latter is proportional to the nominal wage, whereas the entry cost depends on both labor and capital costs. Consequently, an increase in the wage relative

to the rental rate raises the value of entry relative to its cost, inducing additional firm entry.

The same result can also be interpreted through the allocation of mobile capital. Since capital is freely mobile across countries, a higher wage relative to the common rental rate is associated with a larger amount of capital being allocated for use in that country. Given a fixed labor endowment, this additional capital expands the country's productive capacity. In a model with a fixed set of producers, such as Armington (1969), the increase in productive capacity can be absorbed only along the intensive margin, through greater output by existing producers. With endogenous firm entry and exit, however, adjustment can also occur along the extensive margin: the additional capital supports the entry of new firms, increasing the equilibrium measure of firms in that country. The extensive-margin response therefore provides an additional channel through which an increase in the nominal wage is translated into greater domestic production.

In contrast to the $\phi = 1$ case, endogenous firm entry need not amplify the original welfare effect. Because entry is linked to the nominal wage, it can instead introduce an opposing force on welfare. Shocks that place downward pressure on domestic wages, such as increased import competition, reduce firm entry and thus lead to a decline in domestic varieties. The resulting loss of variety further places upward pressure on the price index, partially offsetting the direct gains from lower trade costs. Thus, capital mobility introduces a more complex entry response that can alter the welfare effects of trade liberalization.

2.7.3 General case: $0 \leq \phi \leq 1$

Having considered the two limiting cases, we now return to the general case. The equilibrium measure of firms can be rearranged as

$$N_s = \frac{L_s}{\xi F_s^E} \frac{\sigma - 1}{\sigma \alpha} \left(\frac{W_s}{P_s} \right)^{\phi(1-\xi)} \left(\frac{W_s}{r} \right)^{(1-\phi)(1-\xi)}. \tag{13}$$

The equilibrium measure of firms is therefore determined by the competing effects of changes in the real wage and the nominal wage.

2.8 Comparative statics

Suppose the two countries are symmetric, so that

$$V_{HF} = V_{FH}, \qquad V_{HH} = V_{FF}, \qquad W_H = W_F, \qquad B_H = B_F, \qquad N_H = N_F.$$

I log-linearize the equilibrium conditions around this symmetric baseline to characterize the response to changes in iceberg trade costs. Under symmetry, define the common bilateral trade share as

$$\lambda \equiv \frac{V_{HF}}{\sum_n V_{Hn}} = \frac{V_{FH}}{\sum_n V_{Fn}}.$$

The full system of equations is provided in Appendix L. I now characterize the comparative statics for the two limiting production structures.

2.8.1 Production with labor and intermediate inputs only: $\phi = 1$

In this case, firm entry responds according to

$$\hat{N}_s = (1-\xi)(\hat{W}_s - \hat{P}_s). \tag{14}$$

I consider bilateral and unilateral changes in trade costs in turn.

Bilateral tariff increase. Suppose $\hat{\tau}_{HF} = \hat{\tau}_{FH} \equiv \hat{\tau} > 0$. The equilibrium responses are

$$\begin{pmatrix} \hat{W}_F \\ \hat{P}_H \\ \hat{P}_F \end{pmatrix} = \begin{pmatrix} 0 \\ \frac{\lambda\hat{\tau}}{1-\frac{\sigma}{\sigma-1}(1-\xi)} \\ \frac{\lambda\hat{\tau}}{1-\frac{\sigma}{\sigma-1}(1-\xi)} \end{pmatrix}.$$

The welfare change for domestic agents is thus

$$\hat{W}_H - \hat{P}_H = -\hat{P}_H = -\lambda\hat{\tau}\frac{1}{1-\frac{\sigma}{\sigma-1}\boldsymbol{(1-\xi)}} \tag{15}$$

When $\xi = 1$, the result collapses to the original Melitz-Chaney result, such that the export share is a sufficient statistic for evaluating the welfare loss from an increase in tariffs. When $\xi < 1$, however, welfare depends on the sign of $1 - \frac{\sigma}{\sigma-1}(1-\xi)$

Proposition 1 *If the propagation of price changes through intermediate-input costs is convergent, $\beta \equiv \frac{\sigma}{\sigma-1}(1-\xi) < 1$, then a bilateral increase in trade costs raises the domestic price index.*

Proof. The equilibrium price-index response can be written as

$$\hat{P}_H = \lambda\hat{\tau} + \beta\hat{P}_H, \qquad \beta \equiv \frac{\sigma}{\sigma-1}(1-\xi).$$

The first term is the direct effect of higher trade costs, while the second captures the feedback through intermediate-input costs. When $\beta < 1$, this feedback is convergent, and $\hat{P}_H$ moves in the same direction as $\hat{\tau}$. ■

We consider only the case in which the propagation effect is convergent. As emphasized by Caliendo et al. (2023), the markup $\frac{\sigma}{\sigma-1}$ distorts the allocation of intermediate inputs away from the efficient allocation. Suppose this distortion is eliminated by a first-best producer subsidy, as in Caliendo et al. (2023). The markup term $\frac{\sigma}{\sigma-1}$ then drops out of the intermediate-input channel in $\hat{P}_n$, and the welfare effect reduces to $\frac{\lambda\hat{\tau}}{\xi}$. Since $\xi < 1$, the welfare response to a change in trade costs is larger in magnitude than in the standard Melitz–Chaney model, reflecting the amplification effect discussed in Section 2.7.1.

Unilateral tariff increase. Suppose $\hat{\tau}_{HF} = 0$ and $\hat{\tau}_{FH} \equiv \hat{\tau} > 0$. The equilibrium responses are

$$\begin{pmatrix} \hat{W}_F \\ \hat{P}_H \\ \hat{P}_F \end{pmatrix} = \begin{pmatrix} -\frac{\alpha(1-\lambda)(\sigma\xi-1)}{A}\hat{\tau} \\ \frac{\sigma-1}{\sigma\xi-1}\frac{B}{A}\lambda\hat{\tau} \\ -\frac{\sigma-1}{\sigma\xi-1}\frac{C}{A}\hat{\tau} \end{pmatrix}.$$

where

$$A = (\sigma\xi-1)(2\epsilon(1-\lambda)-1) + 2\lambda(\sigma-1)$$

$$B = \epsilon(1-\lambda)(\sigma\xi-1) + (\sigma-1)\lambda$$

$$C = (\sigma\xi-1)\left(\epsilon(\xi-1+\lambda^2-\lambda\xi)+\alpha+\lambda\right) + \lambda(\alpha-(\sigma-1)\epsilon\xi-(\sigma-1)\lambda)$$

, and the welfare change for domestic agents is thus

$$\hat{W}_H - \hat{P}_H = -\hat{P}_H = -\frac{\sigma - 1}{\sigma\xi - 1}\frac{B}{A}\lambda\hat{\tau} \tag{16}$$

Proposition 2 *If the intermediate-input feedback is convergent, roundabout production does not change the qualitative welfare effects of bilateral and unilateral tariff increases relative to the Melitz–Chaney benchmark.*

Proof. The convergence condition is equivalent to

$$\sigma\xi - 1 > 0.$$

Under this condition, $A > 0$ and $B > 0$. Equation 15 then implies that a bilateral tariff increase reduces welfare, while Equation 16 implies the same for a unilateral tariff increase. Thus, as in the Melitz–Chaney benchmark, both bilateral and unilateral tariff increases reduce welfare. ■

Additionally, similar to the bilateral case, with a first-best correction, roundabout production magnifies the welfare effect of unilateral trade liberalization relative to the standard Melitz (2003) model. See Appendix H for details.

2.8.2 Production with labor and mobile capital only: $\phi = 0$

In this case, firm entry responds to changes in the nominal wage according to

$$\hat{N}_s = (1 - \xi)\hat{W}_s. \tag{17}$$

In the absence of intermediate inputs, the price index P_I enters the remaining equilibrium conditions only through B_I, a monotonic transformation of P_I. It is therefore convenient to characterize the equilibrium system in terms of B_I rather than P_I directly.

Bilateral tariff increase. Suppose $\hat{\tau}_{HF} = \hat{\tau}_{FH} \equiv \hat{\tau} > 0$. The equilibrium responses are

$$\begin{pmatrix} \hat{W}_H \\ \hat{W}_F \\ \hat{B}_H \\ \hat{B}_F \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ (\sigma-1)\lambda\hat{\tau} \\ (\sigma-1)\lambda\hat{\tau} \end{pmatrix}.$$

To recover the corresponding change in the price index, note that

$$\hat{P}_n = \frac{1}{\sigma-1}\left(\hat{B}_n - s_n\hat{W}_n\right),$$

where s_n denotes the labor income share,

$$s_n \equiv \frac{W_n L_n}{W_n L_n + rK_n},$$

which equals ξ in the symmetric benchmark. Since $\hat{W}_H = \hat{W}_F = 0$, it follows that

$$\hat{P}_H = \hat{P}_F = \lambda\hat{\tau}.$$

With capital mobility alone, bilateral tariff increase delivers the same welfare outcome as in the standard Melitz-Chaney model, with the export share remaining a sufficient statistic for evaluating the welfare gains from trade liberalization.

Unilateral tariff increase. Suppose $\hat{\tau}_{HF} = 0$ and $\hat{\tau}_{FH} \equiv \hat{\tau} > 0$. The equilibrium responses are

$$\begin{pmatrix} \hat{W}_H \\ \hat{W}_F \\ \hat{B}_H \\ \hat{B}_F \end{pmatrix} = \begin{pmatrix} \frac{\alpha\lambda(1-\lambda)}{A}\,\hat{\tau} \\ -\frac{\alpha\lambda(1-\lambda)}{A}\,\hat{\tau} \\ (\sigma-1)\frac{\lambda C}{A}\,\hat{\tau} \\ -(\sigma-1)\frac{(1-\lambda)D}{A}\,\hat{\tau} \end{pmatrix}.$$

where

$$A = (1-\xi) + 2\lambda\xi + 4\lambda(1-\lambda)(\epsilon\xi - 1) > 0,$$

$$C = \lambda\big(\xi - (1-2\lambda)(1-\epsilon\xi)\big) + \epsilon\xi,$$

$$D = \lambda\big(\xi - (1-2\lambda)(1-\epsilon\xi)\big).$$

Proposition 3 *A unilateral increase in the Home tariff raises the nominal wage in Home and lowers the nominal wage in Foreign by the same magnitude:*

$$\hat{W}_H = -\hat{W}_F > 0.$$

Proof. See Appendix I.

Proposition 3 shows that a unilateral tariff increase places upward pressure on the domestic wage. Because the wage response is proportional to the change in domestic production, which in turn is proportional to capital employed domestically, the tariff induces a capital inflow. In addition, as discussed in Section 2.7.2, the capital inflow makes the fixed cost of entry relatively cheaper compared with the market value of entry, thereby promoting domestic firm entry and increasing the number of domestic varieties. The increase in domestic varieties, in turn, dampens the tariff-induced increase in the domestic price index. To see this mechanism mathematically, substitute the equilibrium solutions into the log-change expression for the price index.

$$\hat{P}_H = \frac{1}{\sigma - 1}\left(\hat{B}_H - \xi\hat{W}_H\right) = \frac{\lambda}{A}\hat{\tau}F, \tag{18}$$

where

$$F = \left[\epsilon(1-\lambda)\left(1 + 2\lambda - \frac{1}{\sigma}\right) + \lambda\right]\xi + \lambda(2\lambda - 1). \tag{19}$$

Proposition 4 *The sign of the Home price-index response to a unilateral increase in trade costs is determined by a strictly increasing function of ξ. There exists a unique $\xi^* \in (0,1)$ such that*

$$\hat{P}_H \begin{cases} < 0, & \xi < \xi^*, \\ = 0, & \xi = \xi^*, \\ > 0, & \xi > \xi^*. \end{cases}$$

Proof. From Equation 19,

$$\frac{\partial F}{\partial \xi} = \epsilon(1-\lambda)\left(1 + 2\lambda - \frac{1}{\sigma}\right) + \lambda > 0,$$

so F is strictly increasing in ξ. Since $F(0) < 0 < F(1)$, there exists a unique $\xi^* \in (0,1)$ such that $F(\xi^*) = 0$. By Equation 18, $\hat{P}_H$ has the same sign as F, which establishes the result. ■

Corollary 1 *Endogenous firm entry dampens the Home price-index response to a unilateral increase in trade costs and, if sufficiently strong, can reverse its sign.*

Proof. Home firm entry satisfies

$$\hat{N}_H = (1-\xi)\hat{W}_H = (1-\xi)(1-\lambda)\frac{\alpha\lambda}{A}\hat{\tau}.$$

Rearranging the Home price-index response gives

$$\hat{P}_H = \underbrace{\left[\epsilon(1-\lambda)\left(1+2\lambda-\frac{1}{\sigma}\right)\xi+2\lambda^2\right]\frac{\lambda\hat{\tau}}{A}}_{\text{conventional price-increasing effect}} \underbrace{-\frac{\lambda\hat{N}_H}{\alpha(1-\lambda)}}_{\text{entry effect}}. \tag{20}$$

The conventional effect raises the price index, while endogenous entry operates in the opposite direction. Hence, entry dampens the price-index increase and, by Proposition 4, reverses its sign when $\xi < \xi^*$. ■

Corollary 2 *A welfare gain from a unilateral increase in trade costs requires a weaker condition on ξ than a decline in the Home price index. There exists a unique $\xi^{**} > \xi^*$ such that*

$$\hat{P}_H \begin{cases} < \xi\hat{W}_H, & \xi < \xi^{**}, \\ = \xi\hat{W}_H, & \xi = \xi^{**}, \\ > \xi\hat{W}_H, & \xi > \xi^{**}. \end{cases}$$

Hence, for $\xi \in (\xi^*, \xi^{**})$, household welfare rises even though the Home price index rises.

Proof. At $\xi = \xi^*$, Proposition 4 implies $\hat{P}_H = 0$. Since $\hat{W}_H > 0$,

$$\xi^*\hat{W}_H - \hat{P}_H = \xi^*\hat{W}_H > 0.$$

Thus household welfare is still increasing at the price-index cutoff. Since the welfare response crosses zero uniquely at ξ^{**}, it follows that $\xi^{**} > \xi^*$. ■

Appendix M considers two extreme cases that illustrate different features of the mechanism. First, as σ approaches one, the responses of equilibrium variables to changes in trade costs converge to zero, so that the capital-mobility channel vanishes. The small-open-economy limit leads to a different result. As Home becomes negligible relative to the rest of the world, Home trade policy no longer affects Foreign equilibrium variables, and global capital market clearing ceases to restrict the response of Home. The effects of changes in trade costs need not vanish in this case; instead, they are determined by the remaining Home equilibrium conditions and depend on the initial pattern of trade flows.

3 2018 Tariff Policy

To analyze the welfare consequences of the 2018 U.S. tariffs on Chinese imports, I extend the two-country framework to an n-region setting and apply exact hat algebra to characterize the equilibrium responses to changes in trade costs. The full equilibrium system is shown in Appendix N.

The policy is modeled as an increase in tariffs imposed by the 51 U.S. regions (the 50 states and Washington, D.C.) on imports from China, from an average tariff rate of 1.66% to 25%. Chinese retaliation is modeled as a symmetric increase in China's tariffs on imports from the United States to 25%.

3.1 Multi-region calibration

The parameters required to solve the exact-hat equilibrium are

$$\Theta = \left\{\sigma,\ \alpha,\ \xi,\ \phi,\ \gamma_S \in \mathbb{R}^{52},\ \tau_{SI} \in \mathbb{R}^{52\times 52}\right\}.$$

Table 1 summarizes the parameter values and their sources. I take the elasticity of substitution and Pareto shape parameter from Melitz and Redding (2015), and calibrate ξ and ϕ using U.S. economy-wide totals for employee compensation, intermediate inputs, and industry output from the 2017 BEA Use Table (U.S. Bureau of Economic Analysis, 2017).

Parameter	Value	Description	Source
σ	4	Elasticity of substitution	Melitz and Redding (2015)
α	4.25	Pareto shape parameter	Melitz and Redding (2015)
ξ	0.31	Labor share	U.S. Bureau of Economic Analysis (2017)
ϕ	0.64	Intermediate-input share conditional on non-labor inputs	U.S. Bureau of Economic Analysis (2017)

Table 1: Parameter values used in the calibration.

I construct a $(52 \times 52 - 1)$ trade-value matrix V using trade data for 51 U.S. regions and China for the year 2017. State-to-state trade flows within the United States are obtained from the Freight Analysis Framework (FAF).[3] State-level exports to and imports from China are constructed using data from USA Trade Online.[4] China's domestic sales are not directly observed. To complete the trade matrix, I impute China's domestic sales using the relative size of the Chinese and U.S. economies[5]:

$$V_{CHN,CHN} + \sum_{I \neq CHN} V_{CHN,I} = \left(\sum_{S \in \text{US}} \sum_{I} V_{S,I} \right) \frac{GDP_{CHN}}{GDP_{US}} \tag{21}$$

where GDP_{CHN} and GDP_{US} denote China's GDP and aggregate U.S. GDP, respectively, obtained from the World Bank.

To assess the validity of the constructed trade matrix, I compare matrix-implied GDP (row sums) with BEA-reported regional GDP in 2017 and matrix-implied personal consump-

[3]U.S. Department of Transportation, Bureau of Transportation Statistics (BTS), and Federal Highway Administration (FHWA), *Freight Analysis Framework (FAF), Version 5*, 2017–2050. DOI: 10.21949/1529116.

[4]U.S. Census Bureau, *USA Trade Online*, https://usatrade.census.gov/.

[5]Domestic trade flows are commonly constructed as GDP minus exports in the structural gravity literature (e.g., Head and Mayer (2014)). Because my dataset contains only bilateral trade between the United States and China and excludes their trade with other partners, this accounting identity cannot be applied directly. I therefore approximate China's total sales using relative GDP sizes and recover domestic sales as the residual.

tion expenditures (PCE) (column sums) with BEA-reported regional PCE in 2017. Figure A1 presents the comparison excluding China, while Figure A2 includes China. The strong correlation between the values implied by the constructed trade matrix and the corresponding reported statistics suggests that the trade matrix provides a reasonable representation of regional economic activity and consumption patterns, supporting its use in the exact hat algebra counterfactual analysis.

For bilateral iceberg trade costs (tariffs), denoted by τ_{SI}, I assume there are no tariffs on trade between U.S. states. For U.S. exports to China, I use the weighted average Chinese tariff on U.S. goods in 2017 (0.0685), obtained from the World Integrated Trade Solution (WITS), and apply this tariff uniformly across all U.S. states. Similarly, for Chinese exports to the United States, I use the average U.S. tariff on Chinese goods in 2017 (0.0265), also applied uniformly across all U.S. states. Then, I abstract the 2018 tariff policy as the United States raising its tariff on Chinese goods to 0.25 ($\widehat{1+\tau}_{CHN,US} = 1.22$), and I abstract Chinese retaliation as China raising its tariff on U.S. goods to 0.25 ($\widehat{1+\tau}_{US,CHN} = 1.18$).

Finally, for the region-specific capital shares γ_S, the equilibrium conditions imply

$$\gamma_S = \frac{\sum_S V_{SI} - (\xi + \phi(1-\xi)) \sum_I V_{SI}}{(1-\xi)(1-\phi) \sum_S \sum_I V_{SI}}. \tag{22}$$

Thus, γ_S can be computed directly from the trade matrix using the row sums, column sums, and the sum of all entries. Alternatively, one can compute γ_S using regional GDP, regional personal consumption expenditure shares, and aggregate GDP across all regions, since row sums proxy for regional GDP, column sums proxy for personal consumption expenditures, and the sum of all entries proxies for total GDP. The resulting values of γ_S are very similar across the two approaches. Using this calculation, North Dakota and Rhode Island would have slightly negative implied capital endowment shares, with values of -0.0021 and -0.0004, respectively. A negative capital endowment share in the model indicates that residents of the state are net borrowers of capital: rather than owning capital and receiving capital income, they make net payments to capital owners in other regions.

3.2 Counterfactual effects on the number of firms

An important channel in the model is that trade policy affects firm entry. Because firm entry and the creation of new business establishments may take time to materialize, I examine whether the model-predicted changes in the number of firms are associated with subsequent changes in the number of establishments relative to the pre-tariff level. Specifically, I regress the percentage change in the number of establishments across 51 U.S. regions in each subsequent year relative to 2017, obtained from the County Business Patterns (CBP), on the model-predicted percentage change in the number of firms under China's retaliation.[6] The observed change in establishments is defined as

$$\hat{N}_{yr} = \frac{N_{yr} - N_{2017}}{N_{2017}} \times 100\%.$$

Because the counterfactual analysis holds population fixed, I control for the corresponding population change,

$$\hat{L}_{yr} = \frac{L_{yr} - L_{2017}}{L_{2017}} \times 100\%,$$

as well as the pre-period change in the number of establishments between 2013 and 2017,

$$\hat{N}_{13\text{–}17} = \frac{N_{2017} - N_{2013}}{N_{2013}} \times 100\%,$$

to account for pre-existing cross-regional trends in establishment growth.

The results are reported in Table 2. The model-predicted percentage change in the number of firms is positively associated with subsequent establishment growth, with the association strengthening over time. In particular, a 1 percentage point increase in predicted firm entry is associated with a 1.316 percentage point increase in establishments by 2021 and a 1.732 percentage point increase by 2022, relative to 2017.

To assess the roles of mobile capital and traded intermediate inputs separately, I consider the two limiting cases of the model, $\phi = 1$ and $\phi = 0$. When $\phi = 1$, production uses only labor and intermediate inputs. I therefore set the labor share to $\xi = 0.56$, equal to one

[6] I use cumulative changes relative to 2017 rather than annual changes because the model predicts the change between the initial and counterfactual equilibria, rather than a year-to-year change.

Table 2: Model-Predicted Firm Entry and Establishment Growth

	(1) $\hat{N}_{2018}$	(2) $\hat{N}_{2019}$	(3) $\hat{N}_{2020}$	(4) $\hat{N}_{2021}$	(5) $\hat{N}_{2022}$
$\hat{N}_{\text{model}}$	0.121	0.279*	0.682	1.316**	1.732*
	(0.167)	(0.168)	(0.550)	(0.646)	(0.916)
Population growth	0.328	0.289	0.529***	0.854***	0.996***
	(0.311)	(0.227)	(0.200)	(0.182)	(0.193)
Pre-trend	0.196***	0.370***	0.567***	0.637***	0.621***
	(0.054)	(0.056)	(0.105)	(0.139)	(0.206)
Constant	-0.828**	-1.605***	-3.498***	-4.224***	-4.090*
	(0.363)	(0.425)	(1.309)	(1.578)	(2.380)
Observations	51	51	51	51	51
R-squared	0.699	0.824	0.793	0.823	0.841

Notes: Robust standard errors in parentheses, computed using the HC3 estimator. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

minus the intermediate input share reported by U.S. Bureau of Economic Analysis (2017). Table A1 reports the corresponding results. In this case, the model-predicted changes in the number of firms are not significantly associated with observed establishment growth.

When $\phi = 0$, production uses only labor and mobile capital. I set the labor share to $\xi = 0.6$, consistent with U.S. data. Table A2 reports the corresponding results. In contrast to the $\phi = 1$ case, the model-predicted changes in the number of firms are positively and significantly associated with observed establishment growth, although the estimated coefficients are relatively large. This directly reflects the small predicted entry responses in this case, consistent with the fact that, without intermediate inputs, the trade policy shock induces only modest changes in endogenous variables.

As Section 2.7.3 shows, the firm-entry response reflects the balance between the real-wage channel associated with roundabout production and the nominal-wage channel associated with capital mobility. The capital-mobility channel tends to dominate the sign of the entry response because the nominal wage and the price index move in opposite directions, making the change in the real wage naturally smaller in magnitude than the change in the nominal wage. At the same time, roundabout production amplifies the responses of equilibrium

variables to trade-policy changes. This mitigates the extremely large coefficient obtained in the $\phi = 0$ case. Combining roundabout production with capital mobility therefore preserves the direction of the entry response generated by capital mobility while moderating its magnitude, bringing the model's predictions more closely in line with the data.

3.3 Counterfactual welfare effects

Figure 1 shows the cross-regional welfare consequences of the 2018 U.S. tariffs on China, with and without China's retaliation, for pure wage earners, hereafter referred to as workers. Under both scenarios, workers in all U.S. regions experience welfare gains. Figure 2 reports the corresponding welfare effects for pure capital-income earners. Since the rental rate r is chosen as the numeraire, their welfare change is the negative of the change in the price index. The resulting distributional pattern resembles that implied by the Stolper–Samuelson theorem when the United States is viewed as the capital-abundant Home country and China as the labor-abundant Foreign country: U.S. workers and Chinese capital owners gain, while U.S. capital owners and Chinese workers lose.

Figure 3 shows the cross-regional welfare effects for representative households, which earn both labor and capital income. Household welfare therefore depends on the share of rental income in total household income. Households that rely more heavily on rental income benefit less from wage increases. Notably, households in North Dakota and Rhode Island experience the largest welfare gains because their implied capital endowment shares are negative. Overall, households in most U.S. states experience welfare gains.

4 Connection to voting outcomes

Directly assessing whether the model-predicted counterfactual welfare effects are reflected in observed welfare outcomes is challenging, as available measures that could approximate regional welfare are typically observed at an annual frequency and may be affected by various

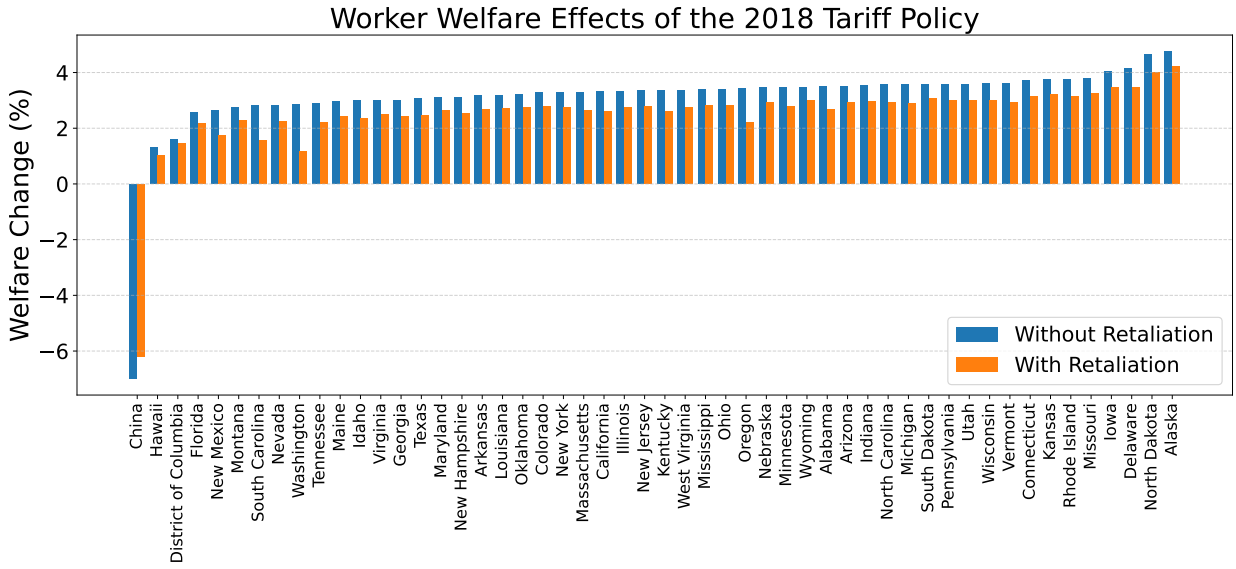


Figure 1: Workers' welfare changes with vs without retaliation.

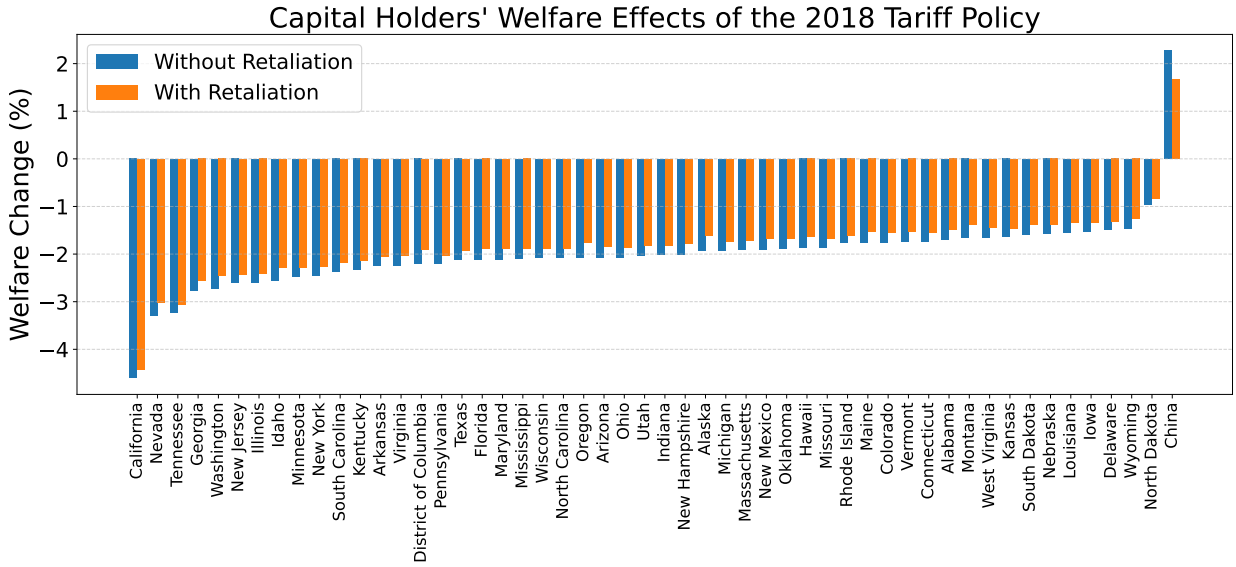


Figure 2: Pure rental income earners' welfare changes with vs without retaliation.

policies and economic shocks. However, tariff policy was a prominent part of Donald Trump's policy agenda during the 2016 presidential campaign. If one is willing to assume that voting decisions partly reflect voters' perceived welfare consequences of the proposed tariff policy, the model-predicted welfare changes can be compared with the change in Republican vote share between the 2012 and 2016 presidential elections.

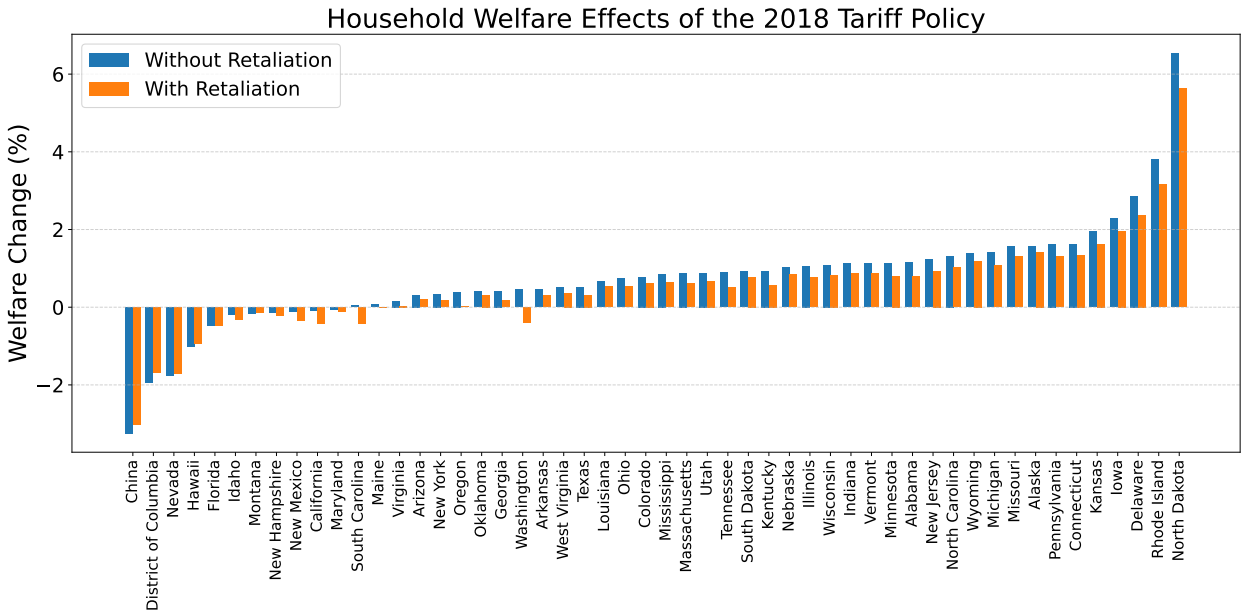


Figure 3: HH welfare changes with vs without retaliation.

The results are reported in Table 3. Model-predicted household welfare changes are positively associated with the change in Republican vote share between the 2012 and 2016 presidential elections. In the general-case specification, a one-percentage-point larger predicted household welfare gain is associated with a 0.876-percentage-point larger increase in the Republican vote share. When China's retaliation is incorporated, the corresponding association increases to 1.028 percentage points. Both estimates are statistically significant at the 1 percent level. The corresponding coefficients for worker welfare are also positive, at 0.639 under the unilateral tariff increase and 0.879 when China's retaliation is included, but are not statistically significant. When there is no mobile capital and production instead features only labor and intermediate inputs, i.e., $\phi = 1$, the relationship between model-predicted worker welfare changes and the change in Republican vote share between 2012 and 2016 becomes negative. In contrast, when production uses only labor and mobile capital, i.e., $\phi = 0$, the positive relationship reappears for household welfare, with a relatively large estimated coefficient. As discussed in the previous section, the large coefficient reflects the relatively small magnitude of the model-predicted welfare changes in the absence of intermediate inputs. Overall, conditional on the assumption that voting decisions partly reflect

Table 3: Republican Vote Share Change (2016–2012) vs. Model-Predicted State Welfare

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
%Δ HH welfare	0.876***							
	(0.228)							
%Δ HH welfare (retaliation)		1.028***						
		(0.259)						
%Δ Worker welfare			0.639					
			(1.102)					
%Δ Worker welfare (retaliation)				0.879				
				(1.020)				
$\phi = 1$: %Δ Worker welfare					-0.156*			
					(0.085)			
$\phi = 1$: %Δ Worker welfare (retaliation)						-0.461		
						(0.281)		
$\phi = 0$: %Δ HH welfare (retaliation)							3.118***	
							(1.051)	
$\phi = 0$: %Δ Worker welfare (retaliation)								3.168
								(3.443)
Constant	-1.351*	-1.234*	-2.732	-2.978	-0.588	-0.659	0.273	-3.308
	(0.706)	(0.683)	(3.437)	(2.500)	(0.678)	(0.676)	(0.760)	(2.624)
Observations	51	51	51	51	51	51	51	51
R-squared	0.054	0.059	0.006	0.012	0.022	0.020	0.054	0.013

Notes: Robust standard errors in parentheses, computed using the HC3 estimator. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

voters' perceived welfare consequences of the proposed tariff policy, the results suggest that regions predicted to experience larger welfare gains tended to exhibit larger increases in Republican vote shares between 2012 and 2016 when capital is mobile, whereas this relationship is reversed in the absence of capital mobility.

Additionally, I regress the change in Republican vote share between the 2016 and 2020 presidential elections on the same model-predicted welfare changes. This exercise tests whether the positive association documented for 2012–2016 reflects a persistent relationship between model-predicted welfare gains and changes in Republican support. The results are reported in Table A3. In contrast to the 2012–2016 results, none of the model-predicted welfare changes are significantly associated with the subsequent change in Republican vote share. This contrast is consistent with the interpretation that the earlier relationship is specific to the electoral shift surrounding the proposed tariff policy. As an additional empirical pattern, Table A5 shows that the model-predicted welfare changes are also significantly cor-

related with the 2016 Republican vote share in levels, except in the specification with only roundabout production but no capital mobility.

5 Conclusion

This paper shows that allowing a productive factor—capital—to move across borders, without relying on the factor-intensity differences central to the Heckscher–Ohlin framework, can alter the direction and consequences of trade-policy-induced capital movements. Combined with endogenous firm entry, this mechanism can also overturn the standard welfare result in the Melitz–Chaney framework that an increase in trade costs reduces welfare.

This paper focuses on tariffs, particularly unilateral tariffs. However, the mechanism is more general: any policy that induces capital inflows can generate similar effects. The central channel is that policies attracting capital inflows make capital relatively cheaper, thereby lowering entry costs relative to the market value of operating firms and encouraging additional firm entry. Capital inflows generate a positive income effect through higher wages, while greater firm entry expands product variety and lowers the domestic price index.

However, the model's assumptions do not fully align with the usual intuition for either type of capital: perfect mobility is difficult to reconcile with physical capital, while the static environment without uncertainty is difficult to reconcile with more liquid financial capital. In this paper, I treat capital as an abstract form of "real money" that can be accumulated and embodied in physical productive assets, such as factories or equipment. Its fundamental value is fixed, although its market price may deviate from this fundamental value over short periods. Relaxing this abstraction by incorporating more realistic features of physical or financial capital could help us better understand the role of capital mobility in shaping the effects of trade policy.

Finally, the mobile factor need not be capital. More generally, the mechanism applies whenever a productive factor is mobile across countries and its reallocation affects the re-

sources available for domestic production and firm entry. The framework could therefore also be extended to labor mobility. In that case, however, migrating workers would bring not only productive capacity but also consumption demand to the destination country, introducing an additional channel absent from the case of mobile capital.

References

Arkolakis, Costas, Arnaud Costinot, and Andrés Rodríguez-Clare. 2012. "New Trade Models, Same Old Gains?" *American Economic Review* 102 (1): 94–130. 10.1257/aer.102.1.94.

Armington, Paul S. 1969. "A Theory of Demand for Products Distinguished by Place of Production (Une théorie de la demande de produits différenciés d'après leur origine) (Una teoría de la demanda de productos distinguiéndolos según el lugar de producción)." *Staff Papers (International Monetary Fund)* 16 (1): 159–178. http://www.jstor.org/stable/3866403.

Autor, David H., David Dorn, and Gordon H. Hanson. 2013. "The China Syndrome: Local Labor Market Effects of Import Competition in the United States." *American Economic Review* 103 (6): 2121–68. 10.1257/aer.103.6.2121.

Bagwell, Kyle and Robert W. Staiger. 1999. "An Economic Theory of GATT." *American Economic Review* 89 (1): 215–248. 10.1257/aer.89.1.215.

Bagwell, Kyle and Robert W. Staiger. 2012. "The economics of trade agreements in the linear Cournot delocation model." *Journal of International Economics* 88 (1): 32–46. https://doi.org/10.1016/j.jinteco.2012.01.013.

Boulhol, Hervé. 2009. "Do capital market and trade liberalization trigger labor market deregulation?" *Journal of International Economics* 77 (2): 223–233. https://doi.org/10.1016/j.jinteco.2008.12.001.

Brainard, S. Lael. 1993. "A Simple Theory of Multinational Corporations and Trade with a Trade-Off Between Proximity and Concentration." Working Paper 4269, National Bureau of Economic Research. 10.3386/w4269.

Caliendo, Lorenzo, Maximiliano Dvorkin, and Fernando Parro. 2019. "Trade and Labor Market Dynamics: General Equilibrium Analysis of the China Trade Shock." *Econometrica* 87 (3): 741–835. https://doi.org/10.3982/ECTA13758.

Caliendo, Lorenzo, Robert C. Feenstra, John Romalis, and Alan M. Taylor. 2023. "A second-best argument for low optimal tariffs on intermediate inputs." *Journal of International Economics* 145: 103824. https://doi.org/10.1016/j.jinteco.2023.103824.

Caliendo, Lorenzo and Fernando Parro. 2021. "Trade Policy." Working Paper 29051, National Bureau of Economic Research. 10.3386/w29051.

Chaney, Thomas. 2008. "Distorted Gravity: The Intensive and Extensive Margins of International Trade." *American Economic Review* 98 (4): 1707–1721. http://www.jstor.org/stable/29730143.

Fajgelbaum, Pablo D, Pinelopi K Goldberg, Patrick J Kennedy, and Amit K Khandelwal. 2020. "The Return to Protectionism." *Quarterly Journal of Economics* 135 (1): 1–55. 10.1093/qje/qjz036.

Galle, Simon, Andrés Rodríguez-Clare, and Moises Yi. 2023. "Slicing the Pie: Quantifying the Aggregate and Distributional Effects of Trade." *Review of Economic Studies* 90 (1): 331–375. 10.1093/restud/rdac020.

Head, Keith and Thierry Mayer. 2014. "Chapter 3 - Gravity Equations: Workhorse,Toolkit, and Cookbook." In *Handbook of International Economics*, vol. 4 of *Handbook of International Economics*, edited by Gita Gopinath, Elhanan Helpman, and Kenneth Rogoff, pp. 131–195. Elsevier. https://doi.org/10.1016/B978-0-444-54314-1.00003-3.

Horstmann, Ignatius J. and James R. Markusen. 1992. "Endogenous market structures in international trade (natura facit saltum)." *Journal of International Economics* 32 (1): 109–129. https://doi.org/10.1016/0022-1996(92)90038-L.

Johnson, Harry G. 1953. "Optimum Tariffs and Retaliation." *Review of Economic Studies* 21 (2): 142–153. 10.2307/2296006.

Krugman, Paul. 1991. "Increasing Returns and Economic Geography." *Journal of Political Economy* 99 (3): 483–499. 10.1086/261763.

Martin, Philippe and Carol Ann Rogers. 1995. "Industrial location and public infrastructure." *Journal of International Economics* 39 (3): 335–351. https://doi.org/10.1016/0022-1996(95)01376-6.

Melitz, Marc J. 2003. "The Impact of Trade on Intra-Industry Reallocations and Aggregate Industry Productivity." *Econometrica* 71 (6): 1695–1725. https://doi.org/10.1111/1468-0262.00467.

Melitz, Marc J. and Gianmarco I. P. Ottaviano. 2008. "Market Size, Trade, and Productivity." *Review of Economic Studies* 75 (1): 295–316. 10.1111/j.1467-937X.2007.00463.x.

Melitz, Marc J. and Stephen J. Redding. 2015. "New Trade Models, New Welfare Implications." *American Economic Review* 105 (3): 1105–46. 10.1257/aer.20130351.

Motta, Massimo. 1992. "Multinational firms and the tariff-jumping argument: A game theoretic analysis with some unconventional conclusions." *European Economic Review* 36 (8): 1557–1571. https://doi.org/10.1016/0014-2921(92)90006-I.

Mundell, Robert A. 1957. "International Trade and Factor Mobility." *American Economic Review* 47 (3): 321–335. http://www.jstor.org/stable/1811242.

U.S. Bureau of Economic Analysis. 2017. "2017 Use Tables." Input-Output Accounts. https://apps.bea.gov/iTable/?reqid=1602&step=2&Categories=Core&isURI=1. U.S. Department of Commerce.

Venables, Anthony J. 1985. "Trade and trade policy with imperfect competition: The case of identical products and free entry." *Journal of International Economics* 19 (1): 1–19. https://doi.org/10.1016/0022-1996(85)90016-9.

A Additional figures

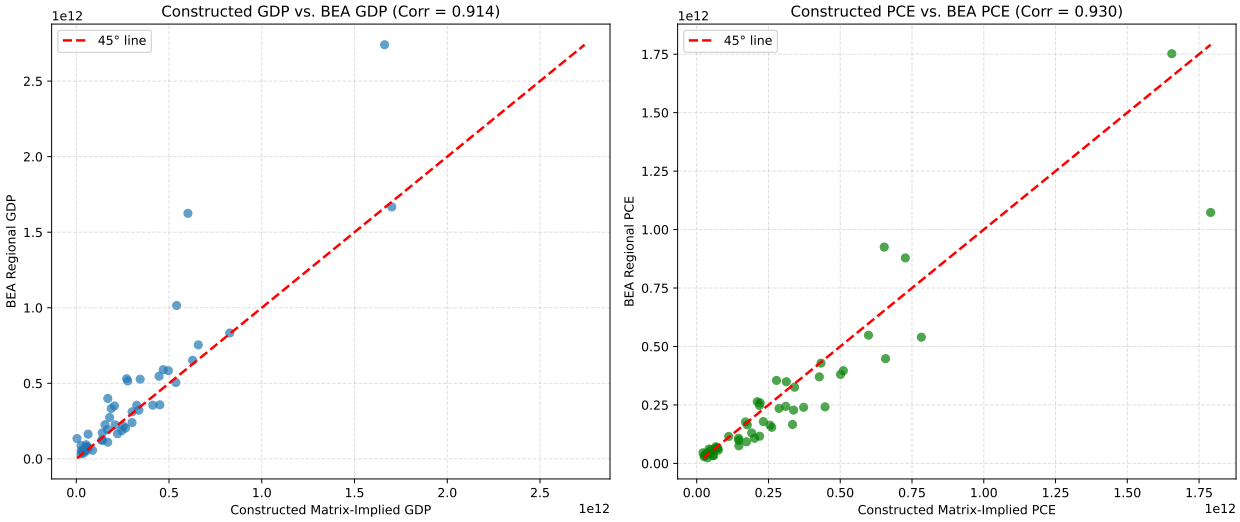


Figure A1: Constructed matrix - implied versus BEA-reported regional values (2017).

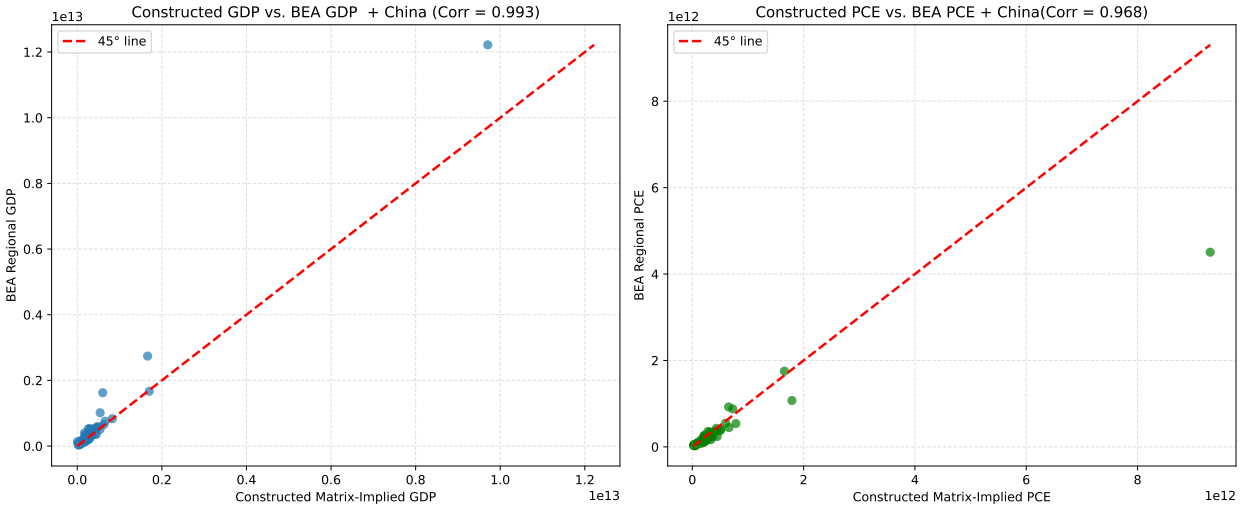


Figure A2: Constructed matrix - implied versus BEA-reported regional values (2017).

B Additional tables

Table A1: $\phi = 1$ Model-Predicted Firm Entry and Establishment Growth

	(1) 2018	(2) 2019	(3) 2020	(4) 2021	(5) 2022
$\hat{N}_{\phi=1}$	-0.003	-0.023	-0.005	-0.044	-0.082
	(0.024)	(0.028)	(0.079)	(0.099)	(0.136)
Population growth	0.312	0.271	0.542***	0.862***	1.000***
	(0.320)	(0.245)	(0.191)	(0.203)	(0.226)
Pre-trend	0.196***	0.373***	0.544***	0.606***	0.588**
	(0.058)	(0.066)	(0.111)	(0.163)	(0.240)
Constant	-0.539***	-0.951***	-1.839***	-1.057**	0.057
	(0.169)	(0.161)	(0.288)	(0.416)	(0.563)
Observations	51	51	51	51	51
R-squared	0.697	0.821	0.786	0.810	0.827

Notes: Robust standard errors in parentheses, computed using the HC3 estimator. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A2: $\phi = 0$ Model-Predicted Firm Entry and Establishment Growth

	(1) $\hat{N}_{2018}$	(2) $\hat{N}_{2019}$	(3) $\hat{N}_{2020}$	(4) $\hat{N}_{2021}$	(5) $\hat{N}_{2022}$
$\hat{N}_{\phi=0}$	0.564	1.099	2.790	5.342**	7.760**
	(0.616)	(0.854)	(1.795)	(2.433)	(3.537)
Population growth	0.352	0.317	0.571***	0.926***	1.077***
	(0.326)	(0.238)	(0.202)	(0.172)	(0.156)
Pre-trend	0.188***	0.350***	0.522***	0.545***	0.492***
	(0.059)	(0.058)	(0.106)	(0.126)	(0.150)
Constant	-1.129*	-2.087**	-4.783**	-6.648**	-8.044**
	(0.602)	(0.936)	(1.901)	(2.615)	(3.904)
Observations	51	51	51	51	51
R-squared	0.701	0.825	0.796	0.826	0.849

Notes: Robust standard errors in parentheses, computed using the HC3 estimator. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A3: Republican Vote Share change (2020-2016) vs. Model-Predicted State Welfare Consequences

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
%Δ HH welfare	-0.214							
	(0.340)							
%Δ HH welfare with retaliation		-0.249						
		(0.381)						
%Δ Worker welfare			-0.492					
			(0.566)					
%Δ Worker welfare with retaliation				-0.502				
				(0.498)				
$\phi = 1$ %Δ Worker welfare					0.031			
					(0.045)			
$\phi = 1$ %Δ Worker welfare with retaliation						0.081		
						(0.142)		
$\phi = 0$ %Δ HH welfare with retaliation							-1.008	
							(0.855)	
$\phi = 0$ %Δ Worker welfare with retaliation								-1.124
								(1.558)
Constant	1.005***	0.974***	2.449	2.171*	0.819***	0.833***	0.537	1.779
	(0.389)	(0.354)	(1.839)	(1.286)	(0.315)	(0.315)	(0.430)	(1.250)
Observations	51	51	51	51	51	51	51	51
R-squared	0.015	0.016	0.016	0.018	0.004	0.003	0.027	0.008

Notes: Robust standard errors in parentheses, computed using the HC3 estimator. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A4: Republican Vote Share change (2024-2020) vs. Model-Predicted State Welfare Consequences

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
%Δ HH welfare	-0.100							
	(0.105)							
%Δ HH welfare with retaliation		-0.113						
		(0.118)						
%Δ Worker welfare			-0.324					
			(0.280)					
%Δ Worker welfare with retaliation				-0.188				
				(0.304)				
$\phi = 1$ %Δ Worker welfare					-0.022			
					(0.033)			
$\phi = 1$ %Δ Worker welfare with retaliation						-0.075		
						(0.104)		
$\phi = 0$ %Δ Worker welfare with retaliation							-0.487	
							(0.454)	
$\phi = 0$ %Δ HH welfare with retaliation								-0.954
								(1.421)
Constant	2.305***	2.289***	3.290***	2.725***	2.226***	2.215***	2.082***	3.030**
	(0.208)	(0.200)	(0.995)	(0.891)	(0.166)	(0.166)	(0.170)	(1.291)
Observations	51	51	51	51	51	51	51	51
R-squared	0.012	0.012	0.025	0.009	0.007	0.009	0.023	0.021

Notes: Robust standard errors in parentheses, computed using the HC3 estimator. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A5: Republican Vote Share vs. Model-Predicted State Welfare Consequences

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
%Δ HH welfare	2.993*							
	(1.747)							
%Δ HH welfare with retaliation		3.477*						
		(1.876)						
%Δ Worker welfare			8.245*					
			(4.278)					
%Δ Worker welfare with retaliation				7.495**				
				(3.389)				
$\phi = 1$ %Δ Worker welfare					-0.695			
					(0.768)			
$\phi = 1$ %Δ Worker welfare with retaliation						-1.893		
						(2.362)		
$\phi = 0$ %Δ HH welfare, with retaliation							13.963**	
							(6.977)	
$\phi = 0$ %Δ Worker welfare, with retaliation								18.443**
								(8.225)
Constant	45.886***	46.311***	21.153	28.290***	48.532***	48.228***	52.387***	32.745***
	(2.533)	(2.254)	(14.776)	(9.777)	(1.690)	(1.854)	(2.154)	(7.512)
Observations	51	51	51	51	51	51	51	51
R-squared	0.102	0.108	0.155	0.138	0.070	0.054	0.176	0.073
Adj. R-squared	0.083	0.090	0.138	0.121	0.052	0.034	0.159	0.054

Notes: Robust standard errors in parentheses, computed using the HC3 estimator. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

C Proof of the symmetric equilibrium solution when $\phi = 0$

This appendix derives the symmetric equilibrium solution for the case without intermediate inputs ($\phi = 0$). Under the assumption of complete symmetry across countries, the equilibrium conditions simplify as follows.

$$\gamma = \frac{1}{2}$$

$$F_J^E = F_L^E = F^E$$

$$F_{JJ} = F_{LL} = F_0$$

$$F_{JL} = F_{LJ} = F_1$$

$$L_J = L_L = L$$

$$\tau_{JL} = \tau_{LJ} = \tau$$

$$\tau_{JJ} = \tau_{LL} = 0$$

Combining the labor market clearing conditions and the free entry conditions gives the expression for the number of firms as a function of wages:

$$N_S = \frac{\sigma - 1}{\sigma\alpha} \frac{W_S^{1-\xi} L}{\xi F^E}. \tag{23}$$

Substituting Equation 23 into the expression for aggregated trade values, Equation **??**, gives:

$$V_{SI} = \frac{(\sigma - 1)\underline{z}^\alpha}{\alpha + 1 - \sigma} \frac{L}{\xi F^E} W_S^{1-\xi\epsilon} B_I^{\frac{\alpha}{\sigma-1}} F_{SI}^{1+\frac{\alpha}{1-\sigma}} (1 + \tau_{SI})^{-\alpha}. \tag{24}$$

Recall that I need to solve for $\{W_J, W_L, B_J, B_L\}$ using two labor market clearing conditions (one for each country), one global capital market clearing condition, and one country's trade balance condition, which are:

$$\sum_I V_{JI} = \frac{W_J L}{\xi},$$

$$\sum_I V_{LI} = \frac{W_L L}{\xi},$$

$$rK_0 = (1 - \xi)(\sum_I V_{JI} + \sum_I V_{LI}),$$

$$W_J L + \gamma r K_0 = V_{JJ} + V_{LJ}.$$

The proportional relationship between V_{JJ} and V_{JL} is:

$$\frac{V_{JJ}}{V_{JL}} = \frac{B_J^{\frac{\alpha}{\sigma-1}} F_{JJ}^{1+\frac{\alpha}{1-\sigma}}}{B_L^{\frac{\alpha}{\sigma-1}} F_{JL}^{1+\frac{\alpha}{1-\sigma}} (1 + \tau_{JL})^{-\alpha}} = \left(\frac{B_J}{B_L}\right)^{\frac{\alpha}{\sigma-1}} C,$$

where

$$C = \left(\frac{F_0}{F_1}\right)^{1+\frac{\alpha}{1-\sigma}} (1 + \tau)^\alpha.$$

Similarly,

$$\frac{V_{LL}}{V_{LJ}} = \left(\frac{B_L}{B_J}\right)^{\frac{\alpha}{\sigma-1}} C.$$

Denote

$$\beta = \left(\frac{B_J}{B_L}\right)^{\frac{\alpha}{\sigma-1}}.$$

Then I have

$$V_{JJ} = \beta C V_{JL}, \tag{25}$$

$$V_{LL} = \frac{C}{\beta} V_{LJ}. \tag{26}$$

At the same time,

$$\frac{V_{JL}}{V_{LJ}} = \frac{W_J^{1-\xi\epsilon}}{W_L^{1-\xi\epsilon}} \frac{B_L^{\frac{\alpha}{\sigma-1}} F_{JL}^{1+\frac{\alpha}{1-\sigma}} (1+\tau_{JL})^{-\alpha}}{B_J^{\frac{\alpha}{\sigma-1}} F_{LJ}^{1+\frac{\alpha}{1-\sigma}} (1+\tau_{LJ})^{-\alpha}} = \left(\frac{W_J}{W_L}\right)^{1-\xi\epsilon} \left(\frac{B_L}{B_J}\right)^{\frac{\alpha}{\sigma-1}}.$$

So I have

$$\frac{V_{JL}}{V_{LJ}} = \left(\frac{W_J}{W_L}\right)^{1-\xi\epsilon} \frac{1}{\beta}. \tag{27}$$

Combining the two labor market clearing conditions gives:

$$\frac{W_J}{W_L} = \frac{V_{JJ} + V_{JL}}{V_{LL} + V_{LJ}} = \frac{V_{JL}(1+C\beta)}{V_{LJ}(1+\frac{C}{\beta})}. \tag{28}$$

Substituting Equation 27 into Equation 28 gives:

$$\frac{W_J}{W_L} = \frac{1}{\beta} \left(\frac{W_J}{W_L}\right)^{1-\xi\epsilon} \frac{1+C\beta}{1+\frac{C}{\beta}}. \tag{29}$$

This further implies

$$\left(\frac{W_J}{W_L}\right)^{\xi\epsilon} = \frac{1+C\beta}{\beta+C}. \tag{30}$$

Now combining Equation 25 and Equation 26 gives:

$$\frac{V_{JJ}}{V_{LL}} = \frac{\beta C V_{JL}}{\frac{C}{\beta} V_{LJ}} = \beta^2 \frac{V_{JL}}{V_{LJ}} = \beta \left(\frac{W_J}{W_L}\right)^{1-\xi\epsilon}. \tag{31}$$

Substituting Equation 30 back gives:

$$\frac{V_{JJ}}{V_{JL}} = \frac{W_J}{W_L} \frac{\beta^2 + C\beta}{1+C\beta}, \tag{32}$$

and

$$\frac{V_{JL}}{V_{LJ}} = \frac{W_J}{W_L} \frac{\beta + C}{\beta + C\beta^2}. \tag{33}$$

Recall that combining the labor market clearing conditions gives:

$$\frac{W_J}{W_L} = \frac{V_{JJ} + V_{JL}}{V_{LL} + V_{LJ}}.$$

Representing V_{JJ} using V_{LL} and V_{JL} using V_{LJ} gives:

$$\frac{W_J}{W_L} = \frac{\frac{W_J}{W_L}\frac{\beta^2+C\beta}{1+C\beta} V_{LL} + \frac{W_J}{W_L}\frac{\beta+C}{\beta+C\beta^2} V_{LJ}}{V_{LL} + V_{LJ}}.$$

This further implies

$$1 = \frac{\frac{\beta^2+C\beta}{1+C\beta} V_{LL} + \frac{\beta+C}{\beta+C\beta^2} V_{LJ}}{V_{LL} + V_{LJ}}. \tag{34}$$

For this equation to hold, it must be that

$$\frac{\beta^2 + C\beta}{1+C\beta} = \frac{\beta + C}{\beta + C\beta^2} = 1, \tag{35}$$

which implies $\beta = 1$, i.e. $B_J = B_L$. From Equation 30, this also implies $W_J = W_L$.

Using $B_J = B_L = B$ and $W_J = W_L = W$, I can solve for

$$W = \frac{L\xi}{2K_0(1-\xi)}$$

and

$$B^{\frac{\alpha}{\sigma-1}} = \left(\frac{L\xi}{2K_0(1-\xi)}\right)^{\xi\epsilon} \frac{\alpha+1-\sigma}{(\sigma-1)\underline{z}^{\alpha}} \frac{F^E}{F_0^{1+\frac{\alpha}{1-\sigma}} + F_1^{1+\frac{\alpha}{1-\sigma}}(1+\tau)^{-\alpha}}.$$

Therefore, the solution of the model under symmetric equilibrium is as follows:

$$\begin{pmatrix} W_J \\ W_L \\ B_J \\ B_L \end{pmatrix} = \begin{pmatrix} \frac{L\xi}{2K_0(1-\xi)} \\ \frac{L\xi}{2K_0(1-\xi)} \\ \left[\left(\frac{L\xi}{2K_0(1-\xi)}\right)^{\xi\epsilon} \frac{\alpha+1-\sigma}{(\sigma-1)\underline{z}^{\alpha}} \frac{F^E}{F_0^{1+\frac{\alpha}{1-\sigma}} + F_1^{1+\frac{\alpha}{1-\sigma}}(1+\tau)^{-\alpha}}\right]^{\frac{\sigma-1}{\alpha}} \\ \left[\left(\frac{L\xi}{2K_0(1-\xi)}\right)^{\xi\epsilon} \frac{\alpha+1-\sigma}{(\sigma-1)\underline{z}^{\alpha}} \frac{F^E}{F_0^{1+\frac{\alpha}{1-\sigma}} + F_1^{1+\frac{\alpha}{1-\sigma}}(1+\tau)^{-\alpha}}\right]^{\frac{\sigma-1}{\alpha}} \end{pmatrix}.$$

D Immobile capital

This appendix considers a variant of the model in Section 2 in which capital is immobile across countries. I show that, under immobile capital, the equilibrium mass of firms is pinned down by exogenous parameters rather than adjusting to endogenous equilibrium variables.

As in Section 2, the total world capital endowment is K_0. Households in country J own a share γ of this endowment, while households in country L own the remaining share $1-\gamma$. The parameters K_0, γ, and $1-\gamma$ are all exogenously given.

D.1 Consumer side

Consumers (households) in destination country $I \in \{J, L\}$ solve the same utility-maximization problem subject to their budget constraint as in Section 2, except that the rental rate of capital is now country specific. In particular, instead of a common rental rate r, consumers in country I face r_I.

Given this modification, the demand for variety ω by consumers in country I is

$$q_I(\omega) = \frac{p_I(\omega)^{-\sigma}}{P_I^{1-\sigma}} \left(r_I K_I + W_I L_I\right).$$

D.2 Firm side

On the firm side, the only modification relative to Section 2 is that the rental rate of capital is country specific. Firms located in country $S \in \{J, L\}$ therefore face the rental rate r_S.

As illustrated in Section 2, with constant source–destination iceberg trade costs, there is a one-to-one mapping between the quantity consumed in destination I, $q_I(\omega)$, and the quantity shipped by a firm with productivity z from source S to destination I, $q_{SI}(z)$.

Given demand for goods shipped to destination I, a firm located in country S requires the following inputs:

$$K_{SI}(z) = (1-\xi)\frac{q_{SI}(z)}{z}\left(\frac{W_S}{r_S}\right)^{\xi}(1+\tau_{SI}),$$

$$L_{SI}(z) = \xi\frac{q_{SI}(z)}{z}\left(\frac{W_S}{r_S}\right)^{\xi-1}(1+\tau_{SI}).$$

Finally, under monopolistic competition, firms set prices as a constant markup over marginal cost:

$$p_{SI}(z) = \frac{\sigma}{\sigma-1}\frac{W_S^{\xi} r_S^{1-\xi}}{z}(1+\tau_{SI}).$$

D.3 Aggregated trade values

The trade value generated by a firm with productivity z located in country S and selling to destination I is

$$V_{SI}(z) = \sigma B_I \left(\frac{W_S^{\xi} r_S^{1-\xi}}{z}(1+\tau_{SI})\right)^{1-\sigma}.$$

Aggregating over all firms in country S that export to market I, total trade values are given by

$$V_{SI} = \frac{\sigma\alpha \underline{z}^{\alpha}}{\alpha+1-\sigma} N_S B_I^{\frac{\alpha}{\sigma-1}} F_{SI}^{1+\frac{\alpha}{1-\sigma}} \left(W_S^{\xi} r_S^{1-\xi}\right)^{1-\epsilon} (1+\tau_{SI})^{-\alpha}. \tag{36}$$

Relative to Equation **??**, the only difference is that the rental rate of capital is now country specific.

D.4 Technology threshold for selling in a market

The productivity threshold for a firm located in country S to profitably sell in destination market I is analogous to Equation 3, except that the rental rate of capital is now country specific. Specifically,

$$z_{\min}^{SI} = \left(\frac{F_{SI}}{B_I}\right)^{\frac{1}{\sigma-1}} \left(W_S^{\xi} r_S^{1-\xi}\right)^{\frac{\sigma}{\sigma-1}} (1+\tau_{SI}).$$

D.5 Equilibrium conditions

With immobile capital, the global capital-market–clearing condition is replaced by country-specific capital-market–clearing conditions, each associated with its own rental rate. Normalizing the wage in country J to one, the endogenous variables are

$$W_L,\ r_J,\ r_L,\ N_J,\ N_L,\ B_J,\ B_L.$$

These seven endogenous variables are pinned down by seven equilibrium conditions. For each country, these include:

1. a free-entry condition,
2. a labor-market–clearing condition, and
3. a capital-market–clearing condition.

In addition, one trade-balance condition closes the model.

The free-entry conditions are

$$\frac{\sigma-1}{\sigma\alpha}\sum_I V_{JI} = N_J F_J^E W_J^{\xi} r_J^{1-\xi}, \tag{37}$$

$$\frac{\sigma-1}{\sigma\alpha}\sum_I V_{LI} = N_L F_L^E W_L^{\xi} r_L^{1-\xi}. \tag{38}$$

Labor-market–clearing conditions are

$$W_J L_J = \xi \sum_I V_{JI}, \tag{39}$$

$$W_L L_L = \xi \sum_I V_{LI}. \tag{40}$$

Capital-market–clearing conditions are

$$r_J K_J = r_J \gamma K_0 = (1-\xi)\sum_I V_{JI}, \tag{41}$$

$$r_L K_L = r_L (1-\gamma) K_0 = (1-\xi)\sum_I V_{LI}. \tag{42}$$

Finally, the trade-balance condition is

$$V_{JL} = V_{LJ}. \tag{43}$$

Notice that combining the country-specific labor-market–clearing condition with the capital-market–clearing condition implies

$$\frac{W_S}{r_S} = \frac{\xi}{1-\xi}\frac{K_S}{L_S}. \tag{44}$$

Equation (44) shows that the relative factor price is pinned down by factor endowments, which are exogenous. As a result, the rental rate r_S can be expressed as a function of the wage W_S, allowing r_S to be eliminated as an independent equilibrium object. The set of endogenous variables can therefore be reduced to

$$\{W_L,\, N_J,\, N_L,\, B_J,\, B_L\}.$$

Next, combining the country-specific free-entry condition with the labor-market–clearing condition yields

$$N_S = \frac{\sigma - 1}{\sigma\alpha} \frac{L_S}{\xi F_S^E} \left(\frac{W_S}{r_S} \right)^{1-\xi}. \tag{45}$$

Substituting Equation (44) into Equation (45) gives

$$N_S = \frac{\sigma - 1}{\sigma\alpha} \frac{1}{\xi F_S^E} \left(\frac{\xi}{1-\xi} \right)^{1-\xi} K_S^{1-\xi} L_S^{\xi}. \tag{46}$$

Equation (46) implies that the equilibrium mass of firms N_S depends solely on exogenous factor endowments and structural parameters. Consequently, N_J and N_L can also be eliminated as equilibrium objects, further reducing the set of endogenous variables to

$$\{W_L,\, B_J,\, B_L\}.$$

Equivalently, after substituting out r_S and N_S, the equilibrium system is isomorphic to one in which labor is the only immobile input, with capital affecting equilibrium outcomes only through the composite term $K_S^{1-\xi} L_S^{\xi}$.

As a result, to determine the remaining endogenous variables $\{W_L, B_J, B_L\}$, it suffices to impose the two country-specific labor-market–clearing conditions together with the trade-balance condition:

$$W_J L_J = \xi \sum_I V_{JI}, \tag{47}$$

$$W_L L_L = \xi \sum_I V_{LI}, \tag{48}$$

$$V_{JL} = V_{LJ}. \tag{49}$$

Number of firms From Equation (45), all variables on the right-hand side are exogenous. In particular, K_S and L_S are fixed country-specific factor endowments, while ξ and F_S^E are structural parameters. Therefore, under immobile capital, the equilibrium mass of firms in country $S \in \{J, L\}$ is pinned down entirely by exogenous primitives and does not respond to endogenous equilibrium outcomes such as wages, rental rates, or market size. This result highlights that it is the interaction between capital usage in production and capital mobility that causes firm entry to respond endogenously to factor prices, as shown in Section 2.

D.6 Comparative statics

Log-differentiate Equation 36 gives:

$$\hat{V}_{SI} = \frac{\alpha}{\sigma - 1}\hat{B}_I + (1-\epsilon)\xi\hat{W}_S + (1-\xi)(1-\epsilon)\hat{r}_S - \alpha\hat{\tau}_{SI}$$

Since $\frac{W_S}{r_S}$ is equal to a constant, I have

$$\hat{W}_S = \hat{r}_S$$

So

$$\hat{V}_{SI} = \frac{\alpha}{\sigma - 1}\hat{B}_I + (1-\epsilon)\hat{W}_S - \alpha\hat{\tau}_{SI} \tag{50}$$

Given this, I can then log-differentiate equilibrium conditions: For country J's labor market:

$$\sum_I V_{SI}\hat{W}_S = \sum_I V_{SI}\hat{V}_{SI}$$

Denote $\lambda_S = \frac{V_{SI}}{\sum_I V_{SI}}$, under symmetry, $\lambda_J = \lambda_L = \lambda$ So I have:

$$\hat{W}_J = \lambda\hat{V}_{JL} + (1-\lambda)\hat{V}_{JJ} \tag{51}$$

Sicne W_J is the numeraire, this furtherly becomes:

$$\lambda\hat{V}_{JL} = (\lambda - 1)\hat{V}_{JJ} \tag{52}$$

For country L:

$$\hat{W}_L = \lambda\hat{V}_{LJ} + (1-\lambda)\hat{V}_{LL} \tag{53}$$

Finally for the trade balance condition

$$\hat{V}_{JL} = \hat{V}_{LJ} \tag{54}$$

Combining them together, the system can be written in matrix form as

$$\begin{pmatrix} 0 & -\frac{\alpha(1-\lambda)}{\sigma-1} & -\frac{\alpha\lambda}{\sigma-1} \\ \epsilon & -\frac{\alpha\lambda}{\sigma-1} & -\frac{\alpha(1-\lambda)}{\sigma-1} \\ 1-\epsilon & \frac{\alpha}{\sigma-1} & -\frac{\alpha}{\sigma-1} \end{pmatrix} \begin{pmatrix} \hat{W}_L \\ \hat{B}_J \\ \hat{B}_L \end{pmatrix} = \begin{pmatrix} -\alpha\lambda\,\hat{\tau}_{JL} \\ -\alpha\lambda\,\hat{\tau}_{LJ} \\ \alpha(\hat{\tau}_{LJ} - \hat{\tau}_{JL}) \end{pmatrix}.$$

Bilateral trade liberalization Suppose trade costs fall symmetrically,

$$\hat{\tau}_{JL} = \hat{\tau}_{LJ} = \hat{\tau} < 0.$$

Solving the system yields

$$\begin{pmatrix} \hat{W}_L \\ \hat{B}_J \\ \hat{B}_L \end{pmatrix} = \begin{pmatrix} 0 \\ \lambda(\sigma-1)\hat{\tau} \\ \lambda(\sigma-1)\hat{\tau} \end{pmatrix}.$$

Using the relationship $-\hat{P}_I = -\hat{B}_I/(\sigma-1)$, both countries experience identical welfare gains equal to

$$-\lambda\hat{\tau}.$$

Hence, the welfare gains from bilateral trade liberalization coincide with those obtained in Section 2.

Unilateral trade liberalization Now consider a unilateral reduction in trade costs from country L to country J,

$$\hat{\tau}_{LJ} = \hat{\tau} < 0, \qquad \hat{\tau}_{JL} = 0.$$

The solution to the system is

$$\begin{pmatrix} \hat{W}_L \\ \hat{B}_J \\ \hat{B}_L \end{pmatrix} = \begin{pmatrix} -\dfrac{\alpha(1-\lambda)\hat{\tau}}{A} \\ (\sigma-1)\dfrac{\lambda B\,\hat{\tau}}{A} \\ -(\sigma-1)\dfrac{(1-\lambda)B\,\hat{\tau}}{A} \end{pmatrix},$$

where $A \equiv 2\epsilon(1-\lambda) + 2\lambda - 1$ and $B \equiv \epsilon(1-\lambda) + \lambda$.

From Equation (51), it is immediate that in this environment trade-cost changes do not affect domestic GDP. In contrast, as shown in Section 2, Paragraph **??**, a unilateral tariff change does alter domestic GDP when capital is mobile. This difference highlights the role of capital mobility in transmitting trade-policy shocks to aggregate production.

The domestic price index change will be simply

$$\hat{P}_J = \frac{1}{\sigma-1}\hat{B}_J \tag{55}$$

where substituting the solution gives

$$\hat{P}_J = -\frac{\lambda}{1-\lambda}\hat{B}_L \tag{56}$$

E Different input factor intensities for fixed costs

This appendix extends the baseline model by allowing fixed costs and variable production to have different factor intensities. Specifically, firms produce final goods using the Cobb–Douglas technology

Suppose that the factor intensity of fixed costs differs from that of variable production. Specifically, firms produce final goods using the Cobb–Douglas technology

$$z\left(\frac{l}{\xi}\right)^{\xi}\left(\frac{k}{1-\xi}\right)^{1-\xi},$$

where $\xi \in (0,1)$ denotes the labor share in variable production.

In contrast, fixed costs are incurred using a different factor intensity. To serve destination I, firms located in country S must incur a fixed cost that uses labor and capital according to

$$\frac{1}{F_{SI}} \left(\frac{l}{\phi}\right)^{\phi} \left(\frac{k}{1-\phi}\right)^{1-\phi},$$

where $\phi \in (0,1)$ denotes the labor share in fixed costs. As a result, the fixed cost of serving market I for firms located in country S is given by

$$F_{SI} W_S^{\phi} r^{1-\phi}.$$

Similarly, the fixed cost of entry for firms located in country S is given by

$$F_S^E W_S^{\phi} r^{1-\phi}.$$

E.1 Consumer side, firm side, and trade values

Allowing fixed costs to have a different factor intensity does not affect variable-cost–related calculations. As a result, consumer demand, firm pricing, and factor demands associated with variable production remain unchanged.

In particular, the quantity demanded of a variety with productivity z shipped from country S to country I is

$$q_{SI}(z) = \frac{\sigma}{\sigma - 1} \frac{W_S^{\xi} r^{1-\xi}}{z} (1 + \tau_{SI}).$$

The firm's factor demands for variable production are therefore unchanged and given by

$$K_{SI}(z) = (1-\xi) \frac{q_{SI}(z)}{z} \left(\frac{W_S}{r}\right)^{\xi} (1 + \tau_{SI}),$$

$$L_{SI}(z) = \xi \frac{q_{SI}(z)}{z} \left(\frac{W_S}{r}\right)^{\xi - 1} (1 + \tau_{SI}).$$

The optimal price charged by a firm from country S selling to country I is

$$p_{SI}(z) = \frac{\sigma}{\sigma - 1} \frac{W_S^{\xi} r^{1-\xi}}{z} (1 + \tau_{SI}).$$

Finally, revenue earned by a firm with productivity draw z selling from S to I is given by

$$V_{SI}(z) = \sigma B_I \left(W_S^{\xi} r^{1-\xi} (1 + \tau_{SI})\right)^{1-\sigma} z^{\sigma - 1}.$$

All expressions above are identical to those in Section 2.

E.2 Technology threshold and aggregate trade value

The first deviation from Section 2 concerns the productivity threshold for firms located in country S to serve market I. With a different factor intensity for fixed costs, the zero-profit cutoff condition is now given by

$$\tilde{\pi}_{SI}\big(z_{SI}^{\min}\big) = F_{SI} W_S^{\phi} r^{1-\phi}.$$

This condition implies the following productivity threshold:

$$z_{SI}^{\min} = \left(\frac{F_{SI}}{B_I}\right)^{\frac{1}{\sigma-1}} W_S^{\frac{\phi}{\sigma-1}+\xi} r^{\frac{1-\phi}{\sigma-1}+1-\xi}(1+\tau_{SI}).$$

Aggregate trade flows from country S to market I are therefore given by

$$V_{SI} = \frac{\sigma \alpha_S \underline{z}_S^{\alpha_S}}{\alpha_S + 1 - \sigma} N_S F_{SI}^{1+\frac{\alpha_S}{1-\sigma}} B_I^{\frac{\alpha_S}{\sigma-1}} (1+\tau_{SI})^{-\alpha_S} W_S^{\phi-\frac{\alpha_S \phi}{\sigma-1}-\alpha_S \xi} r^{1-\phi-\frac{\alpha_S(1-\phi)}{\sigma-1}-\alpha_S(1-\xi)}.$$

E.3 CRS Properties for factor prices and market-clearing conditions

We first verify that the cost function remains homogeneous of degree one in factor prices when fixed costs have a different factor intensity. The total cost incurred by a firm from country S selling to market I can be written as

$$C(q, W_S, r) = F_{SI} W_S^{\phi} r^{1-\phi} + F_S^E W_S^{\phi} r^{1-\phi} + W_S^{\xi} r^{1-\xi} \frac{q_{SI}(z)(1+\tau_{SI})}{z}.$$

It follows immediately that

$$C(q, \lambda W_S, \lambda r) = \lambda C(q, W_S, r),$$

so the cost function remains homogeneous of degree one in factor prices (W_S, r). Equilibrium allocations therefore depend only on relative factor prices.

Turning to market-clearing conditions, labor is used in variable production as well as in the payment of fixed costs for market access and entry. Under the symmetric-country assumption, total labor used by firms from country S to serve destination I can be written as

$$\tilde{L}_{SI} = \frac{\xi(\sigma-1)}{\sigma W_S} V_{SI} + \frac{\phi(\alpha+1-\sigma)}{\sigma \alpha W_S} V_{SI}.$$

Aggregating across destinations and adding entry fixed costs yields total labor demand in country S:

$$L_S = \left[\frac{(\sigma-1)\xi + \phi}{\sigma W_S}\right] \sum_I V_{SI}.$$

Hence, labor-market–clearing conditions can be written compactly as

$$L_J = \frac{(\sigma-1)\xi+\phi}{\sigma W_J}\sum_I V_{JI}, \qquad L_L = \frac{(\sigma-1)\xi+\phi}{\sigma W_L}\sum_I V_{LI}.$$

Similarly, capital is used in variable production and in the payment of fixed costs. The global capital-market–clearing condition is therefore given by

$$K_0 = \frac{(\sigma-1)(1-\xi)+(1-\phi)}{\sigma r}\sum_S\sum_I V_{SI}.$$

Note that when $\phi = \xi$, the market-clearing conditions collapse to those in Section 2.

E.4 Firm entry decisions

Combining the free-entry condition

$$\frac{\sigma-1}{\sigma\alpha}\sum_I V_{SI} = N_S F_S^E W_S^\phi r^{1-\phi}$$

with the labor-market–clearing condition

$$\sigma W_S L_S = ((\sigma-1)\xi+\phi)\sum_I V_{SI}$$

yields the equilibrium mass of firms:

$$N_S = \frac{(\sigma-1)L_S}{\alpha F_S^E\left((\sigma-1)\xi+\phi\right)}\left(\frac{W_S}{r}\right)^{1-\phi}.$$

As in the baseline model, firm entry depends on relative factor prices. A higher rental–wage ratio raises the cost of entry thus reducing the equilibrium mass of firms. Allowing fixed costs to have a different factor intensity alters the elasticity of entry with respect to factor prices but does not change the underlying mechanism.

F Constant fixed costs

This appendix considers the case in which the fixed cost of serving market I from country S, denoted f_{SI}, and the fixed cost of operating (entry), denoted f_S^E, are both constants (i.e., they do not scale with factor prices). Since the rental rate of capital is chosen as the numeraire, these fixed costs are denominated in units of capital and therefore do not vary with equilibrium wages. Equivalently, this specification corresponds to setting the wage intensity parameter ϕ introduced in Appendix E equal to zero. This modification does not affect the household problem or firms' pricing rules, but it alters the export cutoff and the free-entry condition.

Revenue at the firm level. For a firm in S with productivity z selling to market I, revenue is unchanged:

$$V_{SI}(z) = \sigma B_I \Big(W_S^{\xi} r^{1-\xi} (1+\tau_{SI}) \Big)^{1-\sigma} z^{\sigma-1}.$$

Market-access cutoff. Let $\pi_{SI}(z)$ denote operating profits (before paying the market-specific fixed cost). With CES demand and monopolistic competition,

$$\pi_{SI}(z) = \frac{1}{\sigma} V_{SI}(z).$$

The cutoff productivity $z_{\min}^{SI}$ is pinned down by the zero-profit condition

$$\pi_{SI}(z_{\min}^{SI}) = f_{SI},$$

which implies

$$z_{\min}^{SI} = \left(\frac{\sigma f_{SI}}{B_I} \right)^{\frac{1}{\sigma-1}} W_S^{\xi} r^{1-\xi} (1+\tau_{SI}).$$

Aggregate trade value. The aggregate value of sales from S to I is

$$V_{SI} = N_S \int_{z_{\min}^{SI}}^{\infty} V_{SI}(z) f^S(z)\, dz = \frac{\sigma \alpha_S}{\alpha_S + 1 - \sigma} N_S \, \underline{z}_S^{\alpha_S} \, B_I \Big(W_S^{\xi} r^{1-\xi} (1+\tau_{SI}) \Big)^{1-\sigma} \left(z_{\min}^{SI} \right)^{\sigma-1-\alpha_S}.$$

Substituting the cutoff yields

$$V_{SI} = N_S \frac{\sigma \alpha_S \underline{z}^{\alpha_S}}{\alpha_S + 1 - \sigma} f_{SI}^{1+\frac{\alpha_S}{1-\sigma}} B_I^{\frac{\alpha_S}{\sigma-1}} (\Big(W_S^{\xi} r^{1-\xi} (1+\tau_{SI}) \Big)^{-\alpha_S}$$

Profits net of market-access fixed costs. Profits net of the market-specific fixed cost are

$$\tilde{\pi}_{SI}(z) = \pi_{SI}(z) - f_{SI} = \frac{V_{SI}(z)}{\sigma} - f_{SI}.$$

Aggregating across active firms,

$$\tilde{\pi}_{SI} = N_S \int_{z_{\min}^{SI}}^{\infty} \tilde{\pi}_{SI}(z) f^S(z)\, dz = \frac{\sigma - 1}{\sigma \alpha_S} V_{SI},$$

where the last equality uses the Pareto assumption and the cutoff defined by $\pi_{SI}(z_{\min}^{SI}) = f_{SI}$.

Free-entry condition. Let total expected operating profits net of market-specific fixed costs be $\sum_I \tilde{\pi}_{SI}$. With constant entry cost f_S^E, the free-entry condition is

$$\sum_I \tilde{\pi}_{SI} = N_S f_S^E, \qquad \text{equivalently} \qquad \sum_I \frac{\sigma - 1}{\sigma \alpha_S} V_{SI} = N_S f_S^E.$$

Labor market clearing. Labor is used only in variable production. Variable labor demand for a firm with productivity z serving I is proportional to revenue:

$$W_S L_{SI}(z) = \xi \frac{\sigma - 1}{\sigma} V_{SI}(z).$$

Aggregating across destinations and active firms yields

$$W_S L_S = \sum_I N_S \int_{z_{\min}^{SI}}^{\infty} W_S L_{SI}(z) f^S(z)\, dz = \xi \frac{\sigma - 1}{\sigma} \sum_I V_{SI}.$$

Combining labor market clearing with the free-entry condition gives an expression for the equilibrium mass of entrants:

$$N_S = \frac{\sigma - 1}{\sigma \alpha_S} \frac{\sum_I V_{SI}}{f_S^E} = \frac{W_S L_S}{\xi \alpha_S f_S^E}.$$

Takeaway. This appendix highlights a general implication of combining heterogeneous-firm free entry with an integrated capital market. Regardless of whether fixed costs are specified in units of factors (e.g., $W_S^{\xi} r^{1-\xi}$) or as constants, the equilibrium mass of entrants N_S is no longer pinned down solely by endowments and parameters.

G Analysis of the elasticity of GDP with respect to wage channel in response to a tariff change

This appendix examines how capital mobility and endogenous firm entry affect the elasticity of the equilibrium response. In an Armington-style model, regardless of capital mobility, the elasticity is

$$1 - \sigma < 0.$$

Chaney (2008) shows that within the Melitz (2003) framework, assuming a Pareto distribution of firm productivity and no firm entry or exit, the elasticity takes the form

$$-\alpha < 0.$$

If firm entry and exit are allowed within the Chaney (2008) framework, the elasticity becomes

$$1 - \epsilon < 0.$$

How the difference arises To clarify the source of these differences, note first that the price index is a function of domestic and foreign unit factor prices. With the rental rate r chosen as the numeraire, unit factor prices depend solely on country-specific wages. Consequently, all equilibrium adjustments are ultimately driven by wage changes.

However, the log-differential of the price index is not linear in the log-differentials of wages. I therefore separate the effect of wage changes from the effect operating through the price index. When discussing the influence of wages on trade values, I refer to the *direct* effect of wages holding the price index fixed, as opposed to the *indirect* effects transmitted through changes in the price index (or, equivalently, B_I).

To build intuition, I begin at the firm level. Consider a firm whose productivity z is sufficiently high that its market participation decisions are unaffected by a potential tariff change. Log-differentiating Equation 1 yields

$$\hat{V}_{SI}(z) = \xi(1-\sigma)\hat{W}_S + (1-\xi)(1-\sigma)\hat{r} + \hat{B}_I + (1-\sigma)\hat{\tau}_{SI}. \tag{57}$$

When capital is immobile, the rental rate is country-specific. As shown in Appendix D, relative factor prices are pinned down by fixed factor endowments, implying $\hat{r}_S = \hat{W}_S$. Substituting this condition yields

$$\hat{V}_{SI}(z) = (1-\sigma)\hat{W}_S + \hat{B}_I + (1-\sigma)\hat{\tau}_{SI}. \tag{58}$$

Holding wages fixed, firm-level trade values admit a partial-equilibrium interpretation. Demand in market I is downward sloping, while supply is perfectly elastic at the constant-markup price implied by marginal cost. Firm-level sales are therefore determined by the intersection of a downward-sloping demand curve and a horizontal supply curve. Trade values reflect the resulting trade-off between prices and quantities as wages change.

In this case, the elasticity of firm-level sales with respect to wages is $1-\sigma$, exactly as in the Armington (1969) model. In the absence of heterogeneous-firm entry and exit, this firm can be interpreted as a representative firm for the country, so its sales elasticity coincides with the aggregate elasticity, replicating the Armington result.

If we go one step further and allow for heterogeneous firms, the relevant object becomes aggregate trade values. Log-differentiating Equation **??** yields

$$\hat{V}_{SI} = \hat{N}_S + \hat{B}_I + \xi(1-\sigma)\hat{W}_S + (1-\xi)(1-\sigma)\hat{r} + (1-\sigma)\hat{\tau}_{SI} + (\sigma-\alpha-1)\hat{Z}^{SI}_{\min}. \tag{59}$$

Under the assumption that the fixed cost is a constant instead of responding to the factor price, as shown in Chaney (2008), this implies

$$\hat{Z}^{SI}_{\min} = \xi\hat{W}_S + (1-\xi)\hat{r} + \frac{1}{1-\sigma}\hat{B}_I + \hat{\tau}_{SI}. \tag{60}$$

Factors are immobile so that $\hat{r}_S = \hat{W}_S$, and manually fixing the entry of firms (as Chaney (2008) did), aggregate trade values reduce to

$$\hat{V}_{SI} = -\alpha\hat{W}_S + \frac{\alpha}{\sigma-1}\hat{B}_I - \alpha\hat{\tau}_{SI}, \tag{61}$$

which replicates the elasticity result in Chaney (2008).

If we further allow for free entry and exit, and let the fixed cost respond to factor prices,

then

$$\hat{Z}_{\min}^{SI} = \frac{1}{1-\sigma}\hat{B}_I + \frac{\sigma}{\sigma-1}\big(\xi\hat{W}_S + (1-\xi)\hat{r}\big) + \hat{\tau}_{SI}. \tag{62}$$

Substituting Equation (62) into Equation (59) yields

$$\hat{V}_{SI} = \hat{N}_S + \frac{\alpha}{1-\sigma}\hat{B}_I + \xi\left(1 - \frac{\sigma\alpha}{\sigma-1}\right)\hat{W}_S + (1-\xi)\left(1 - \frac{\sigma\alpha}{\sigma-1}\right)\hat{r} - \alpha\hat{\tau}_{SI}. \tag{63}$$

Denote $\frac{\sigma\alpha}{\sigma-1} \equiv \epsilon$. Since $\hat{N}_S = 0$, the wage elasticity is $\xi(1-\epsilon)$.

Finally, if capital is immobile, then $\hat{N}_S = (1-\xi)\hat{W}_S$, and the elasticity becomes

$$(1-\xi) + \xi - \xi\epsilon = 1 - \xi\epsilon,$$

reflecting the combined intensive- and extensive-margin responses.

H Amplifying effect of roundabout production under a unilateral tariff

This appendix examines how roundabout production amplifies the equilibrium responses to a unilateral tariff. Suppose that the markup distortion on intermediate inputs is fully offset, so that the effective markup is equal to one. Under this assumption, the intermediate-input term

$$1 - \frac{\sigma}{\sigma-1}(1-\xi)$$

simplifies to

$$1 - (1-\xi) = \xi.$$

Equivalently, I have:

$$\frac{\sigma\xi - 1}{\sigma - 1} = \xi$$

Hence, the welfare effect of unilateral trade liberalization in the home country becomes

$$\hat{W}_H - \hat{P}_H = -\hat{P}_H = \frac{1}{\xi}\frac{\tilde{B}}{\tilde{A}}\lambda\hat{\tau}$$

where given $A = (\sigma\xi - 1)(2\epsilon(1-\lambda) - 1) + 2\lambda(\sigma - 1)$ and $B = \epsilon(1-\lambda)(\sigma\xi - 1) + (\sigma - 1)\lambda$

$$\tilde{A} = (\sigma - 1)\xi(2\epsilon(1-\lambda) - 1) + 2\lambda(\sigma - 1))$$

$$\tilde{B} = \epsilon(1-\lambda)(\sigma - 1)\xi + (\sigma - 1)\lambda$$

So

$$\hat{W}_H - \hat{P}_H = -\hat{P}_H = \frac{\epsilon(1-\lambda) + \frac{\lambda}{\xi}}{\xi\left(2\epsilon(1-\lambda) - 1\right) + 2\lambda}\lambda\hat{\tau}.$$

In the standard Melitz (2003) model, the corresponding welfare effect is

$$-\hat{P}_H^{\text{Melitz}} = \frac{\epsilon(1-\lambda)+\lambda}{2\epsilon(1-\lambda)-1+2\lambda}\lambda\hat{\tau},$$

as shown in Appendix D.6.

For $\xi < 1$, the roundabout-production economy generates a larger welfare response to unilateral trade liberalization than the standard Melitz (2003) model. The amplification arises from endogenous firm entry: the equilibrium mass of firms moves in the same direction as the real wage, thereby reinforcing the initial welfare response. When $\xi = 1$, the roundabout-production channel disappears and the expression collapses to the standard Melitz (2003) case.

I Proof of Proposition 3

This appendix provides the proof of Proposition 3.

Proof. We first establish the sign of A. It follows that $\epsilon = \frac{\sigma\alpha}{\sigma-1} > \sigma > 1$. Using this inequality, $A = (1-\xi) + 2\lambda\xi + 4\lambda(1-\lambda)(\epsilon\xi - 1)$, can be bounded from below as

$$A > (1-\xi)+2\lambda\xi+4\lambda(1-\lambda)(\xi-1) = (1-\xi)(1-4\lambda(1-\lambda))+2\lambda\xi = (1-\xi)(1-2\lambda)^2+2\lambda\xi > 0.$$

Therefore,

$$A > 0.$$

Having established that $A > 0$, the result in Proposition 3 follows directly.

The Home wage response is given by

$$\hat{W}_H = \frac{\alpha\lambda(1-\lambda)}{A}\hat{\tau}.$$

For $\lambda \in (0,1)$ and $\alpha > 0$, the numerator of the coefficient is strictly positive. Since $A > 0$ whenever $\lambda < 1/2$, a unilateral increase in the Home tariff, $\hat{\tau} > 0$, implies

$$\hat{W}_H > 0.$$

Moreover, the capital-market-clearing condition requires the wage changes in the two countries to be equal in magnitude and opposite in sign. Hence,

$$\hat{W}_F = -\hat{W}_H < 0.$$

■

J Mobile Capital Under Armington (1969) Setup

In this section, I show that without the heterogeneous-firm assumption, a capital inflow can't result in a welfare gain. The main intuition is that under a homogeneous-firm setup, capital inflow can increase domestic production and domestic wages, thus generate positive income effect, which is in magnitude smaller than the direct tariff change on the price index.

To illustrate this more specifically, I show the full derivation under the Armington (1969) setup. With only two countries and thus two goods in the world. Consumers in both countries derive utility from consuming a composite of the two goods.

For consumers in country J, they solve the following utility maximization problem:

$$\max_{q_{JJ},q_{JL}} \left(q_{JJ}^{\frac{\sigma-1}{\sigma}} + q_{JL}^{\frac{\sigma-1}{\sigma}} \right) \quad \text{s.t.} \quad p_J q_{JJ} + (1+\tau_{JL}) p_L q_{JL} = Y_J.$$

Consumers in country J demand goods produced in country J in the quantity

$$q_{JJ} = \frac{p_J^{-\sigma}}{P_J^{1-\sigma}} Y_J$$

and goods produced in country L in the quantity

$$q_{JL} = \frac{((1+\tau_{JL})p_L)^{-\sigma}}{P_J^{1-\sigma}} Y_J.$$

Producers in country J receive demand from both countries J and L, such that

$$Q_J = \frac{p_J^{-\sigma}}{P_J^{1-\sigma}} Y_J + \frac{((1+\tau_{LJ})p_J)^{-\sigma}}{P_L^{1-\sigma}} Y_L.$$

Producers then solve a cost minimization problem to determine the factor inputs required for production. The optimal demands for labor and capital are therefore

$$L_J = \xi \frac{Q_J}{z_J} \left(\frac{w_J}{r} \right)^{\xi-1}$$

$$K_J = (1-\xi) \frac{Q_J}{z_J} \left(\frac{w_J}{r} \right)^{\xi}$$

Under monopolistic competition, firms set prices as a constant markup over marginal cost. Therefore,

$$p_J = \frac{\sigma}{\sigma-1} \frac{w_J^{\xi} r^{1-\xi}}{z_J}$$

$$p_L = \frac{\sigma}{\sigma-1} \frac{w_L^{\xi} r^{1-\xi}}{z_L}.$$

J.1 Equilibrium conditions

Thus there are two labor market clearing conditions as:

$$\frac{w_J L_J}{\xi} = \sum_I V_{JI}$$

$$\frac{w_L L_L}{\xi} = \sum_I V_{LI}$$

one capital market clearing condition:

$$\frac{rK_0}{1-\xi} = \sum_I V_{JI} + \sum_I V_{LI}$$

and one trade balance condition:

$$w_J L_J + \gamma r K_0 = V_{JJ} + V_{LJ}$$

with the trade values be in the form of:

$$V_{SI} = \sigma(\frac{W_S^{\xi} r^{1-\xi}(1+\tau_{SI})}{z_S})^{1-\sigma} B_I$$

where $B_I = \frac{w_I L_I + r K_I}{P_I^{1-\sigma}}$

J.2 Comparative statics

Log-differentiating the equilibrium conditions gives

$$\begin{pmatrix} (1-\xi+\sigma\xi)\sum_I V_{JI} & 0 & -V_{JJ} & -V_{JL} \\ 0 & (1-\xi+\sigma\xi)\sum_I V_{LI} & -V_{LJ} & -V_{LL} \\ \sum_I V_{JI} & \sum_I V_{LI} & 0 & 0 \\ \xi(\sum_I V_{JI} - (1-\sigma)V_{JJ}) & \xi(\sigma-1)V_{LJ} & -(V_{JJ}+V_{LJ}) & 0 \end{pmatrix} \begin{pmatrix} \hat{W}_J \\ \hat{W}_L \\ \hat{B}_J \\ \hat{B}_L \end{pmatrix} = \begin{pmatrix} -(\sigma-1)V_{JL}\hat{\tau}_{JL} \\ -(\sigma-1)V_{LJ}\hat{\tau}_{LJ} \\ 0 \\ -(\sigma-1)V_{LJ}\hat{\tau}_{LJ} \end{pmatrix}.$$

Under the symmetric-country assumption, denote the export share by λ. The system simplifies to

$$\begin{pmatrix} (1-\xi+\sigma\xi) & 0 & -(1-\lambda) & -\lambda \\ 0 & (1-\xi+\sigma\xi) & -\lambda & -(1-\lambda) \\ 1 & 1 & 0 & 0 \\ \xi(1-(1-\sigma)(1-\lambda)) & \xi(\sigma-1)\lambda & -1 & 0 \end{pmatrix} \begin{pmatrix} \hat{W}_J \\ \hat{W}_L \\ \hat{B}_J \\ \hat{B}_L \end{pmatrix} = \begin{pmatrix} -(\sigma-1)\lambda\hat{\tau}_{JL} \\ -(\sigma-1)\lambda\hat{\tau}_{LJ} \\ 0 \\ -(\sigma-1)\lambda\hat{\tau}_{LJ} \end{pmatrix}.$$

J.3 Bilateral trade liberalization

When both countries reduce tariffs by the same amount such that $\hat{\tau}_{JL} = \hat{\tau}_{LJ} = \hat{\tau} < 0$, we obtain

$$\begin{pmatrix} \hat{W}_J \\ \hat{W}_L \\ \hat{B}_J \\ \hat{B}_L \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \lambda(\sigma-1)\hat{\tau} \\ \lambda(\sigma-1)\hat{\tau} \end{pmatrix},$$

which is the same result as in Section 2.

J.4 Unilateral trade liberalization

Suppose only the domestic country reduces its tariff, such that $\hat{\tau}_{LJ} = \hat{\tau} < 0$ and $\hat{\tau}_{JL} = 0$. The solution becomes

$$\begin{pmatrix} \hat{W}_J \\ \hat{W}_L \\ \hat{B}_J \\ \hat{B}_L \end{pmatrix} = \begin{pmatrix} \frac{\lambda\hat{\tau}}{A}(\sigma-1)(1-\lambda) \\ -\frac{\lambda\hat{\tau}}{A}(\sigma-1)(1-\lambda) \\ \frac{\lambda\hat{\tau}}{A}(\sigma-1)(1+\xi(\sigma-1)+\xi\lambda) \\ -\frac{\lambda\hat{\tau}}{A}(\sigma-1)\xi((1-2\lambda)(\sigma(1-\lambda)+\lambda)+\lambda) \end{pmatrix},$$

where

$$A = 1 - \xi + 2\lambda\xi(2\lambda - 1 + 2\sigma - 2\lambda\sigma).$$

Given an export share $\lambda < \frac{1}{2}$, it follows that $A > 0$.

The change in the price index is

$$\hat{P}_I = \frac{1}{\sigma-1}\big(\hat{B}_I - s_I\,\hat{W}_I\big),$$

where $s_I = \xi$ under the symmetric setup. Therefore,

$$\hat{P}_J = \frac{\lambda\hat{\tau}}{A}\big(1 - \xi + \xi(\sigma - 1 + 2\lambda)\big) < 0.$$

Thus, unilateral trade liberalization unambiguously reduces the domestic price index.

Note that

$$\hat{W}_J = \hat{V}_J = \hat{\tilde{K}}_J < 0,$$

where $\hat{\tilde{K}}_J$ denotes the amount of capital used by country J for production. Hence, unilateral liberalization leads country J to lose capital.

However, the change in domestic welfare is

$$\xi\hat{W}_J - \hat{P}_J = \frac{\lambda\hat{\tau}}{A}\big(\xi((\sigma-1)(1-\lambda)+1-(\sigma-1+2\lambda))-1\big).$$

Since

$$\xi((\sigma-1)(1-\lambda)+1-(\sigma-1+2\lambda))-1<(\sigma-1)(1-\lambda)+1-(\sigma-1+2\lambda)-1=-(1+\sigma)\lambda<0,$$

we obtain

$$\xi\hat{W}_J-\hat{P}_J>0.$$

Therefore, unilateral trade liberalization raises domestic welfare, while a unilateral tariff increase would unambiguously reduce welfare.

Compared with Section 2, this result shows that, without the extensive-margin effect, a unilateral tariff increase under the symmetric-country setup cannot generate welfare gains, even though it attracts capital inflows. Thus, the standard result that trade liberalization is welfare-improving continues to hold even in the presence of capital mobility.

K Wage–entry linkage through average entry costs

This appendix shows that the wage–entry linkage arises from the interaction between capital usage and capital mobility through average entry costs.

When firm entry is endogenously determined by the free-entry condition, two general implications arise. First, the number of firms is given by

$$N=\frac{\text{total profit}}{\text{fixed cost}},$$

which in this paper takes the form

$$N_S=\frac{\sum_I\tilde{\pi}_{SI}}{F_S^E W_S^{\xi}r^{1-\xi}}.$$

Second, while the cost function remains homogeneous of degree one in factor prices, it is no longer homogeneous of degree one in quantities due to the presence of fixed costs.

Since factor prices enter the cost function homothetically, and total profit is proportional to cost, the number of firms is determined by relative factor prices, which in this paper are summarized by the wage–rental ratio $\frac{W_I}{r}$.

When all factors are cleared domestically (as in Melitz (2003)), relative factor prices are pinned down by domestic endowments and model parameters. As shown in Appendix D, with two immobile factors,

$$\frac{r_S}{W_S}=\frac{1-\xi}{\xi}\frac{L_S}{K_S},\tag{64}$$

where ξ, K_S, and L_S are all exogenous.

However, when one factor is mobile, the two factor prices are no longer perfectly linked,

such that

$$\frac{r}{W_S} = \frac{1-\xi}{\xi}\frac{L_S}{K_0}\frac{\sum_S \sum_I V_{SI}}{\sum_I V_{SI}}. \tag{65}$$

Here, ξ, K_0, and L_S are exogenous parameters, while $\frac{\sum_I V_{SI}}{\sum_S \sum_I V_{SI}}$ is endogenously determined and represents country S's share of world revenue. From the labor market clearing conditions (Equations 7 and 8), this share can be written as

$$\frac{\sum_I V_{SI}}{\sum_S \sum_I V_{SI}} = \frac{W_S L_S}{\sum_S W_S L_S}. \tag{66}$$

In the symmetric two-country case, this expression reduces to

$$\frac{r}{W_J} = \frac{1-\xi}{\xi}\frac{L_S}{K_0}\left(1 + \frac{W_L}{W_J}\right). \tag{67}$$

Equation (67) shows that the rental–wage ratio is decreasing in country J's wage relative to country L's wage. When W_J/W_L increases, country J captures a larger share of world revenue, which lowers its relative rental price. Intuitively, higher wages are associated with a larger domestic market and higher GDP, which attracts a greater share of mobile capital for production in country J, regardless of capital ownership. This increase in effective capital availability reduces the relative rental rate, making firm entry cheaper and leading to a larger equilibrium mass of firms in country J.

L Log-linearized system of equations

This appendix presents the log-linearized equilibrium system used to characterize the responses of wages, capital allocation, firm entry, and welfare to changes in trade policy.

Labor market clearing conditions. Log-differentiating the labor market clearing conditions yields

$$\sum_n V_{sn}\,\hat{W}_s = \sum_n V_{sn}\,\hat{V}_{sn}. \tag{68}$$

Capital market clearing condition. Log-linearizing the capital market clearing condition in Equation 9 yields

$$\sum_n V_{Hn}\hat{V}_{Hn} + \sum_n V_{Fn}\hat{V}_{Fn} = 0. \tag{69}$$

Using Equation 68, Equation 69 can be rewritten as

$$\sum_n V_{Hn}\hat{W}_H + \sum_n V_{Fn}\hat{W}_F = 0. \tag{70}$$

Thus, equilibrium wage changes in the two countries must move in opposite directions. Under the symmetric baseline,

$$\sum_n V_{Hn} = \sum_n V_{Fn},$$

so that

$$\hat{W}_H = -\hat{W}_F. \tag{71}$$

Trade balance conditions. Log-differentiating the trade balance condition for country H yields

$$\big(\phi(1-\xi)+\xi\big)\sum_n V_{Hn}\hat{W}_H = V_{HH}\hat{V}_{HH} + V_{FH}\hat{V}_{FH}. \tag{72}$$

Under the symmetric baseline, define the common bilateral trade share as

$$\lambda \equiv \frac{V_{HF}}{\sum_n V_{Hn}} = \frac{V_{FH}}{\sum_n V_{Fn}}. \tag{73}$$

L.1 Log-differentiated bilateral trade values

When $\phi = 1$, the log-differentiated bilateral trade value is given by

$$\hat{V}_{sn} = (1-\xi\epsilon)\hat{W}_s + \frac{\alpha}{\sigma-1}\hat{W}_n - (1-\xi)\epsilon\hat{P}_s + \alpha\hat{P}_n - \alpha\hat{\tau}_{sn}. \tag{74}$$

When $\phi = 0$, the log-differentiated bilateral trade value is given by

$$\hat{V}_{sn} = (1-\xi\epsilon)\hat{W}_s + \frac{\alpha s_n}{\sigma-1}\hat{W}_n + \alpha\hat{P}_n - \alpha\hat{\tau}_{sn}, \tag{75}$$

where

$$s_n \equiv \frac{W_n L_n}{W_n L_n + r K_n}$$

denotes the share of labor income in total income.

More generally, when $0 < \phi < 1$, the log-differentiated bilateral trade value is given by

$$\hat{V}_{sn} = (1-\xi\epsilon)\hat{W}_s + \frac{\alpha s_n}{\sigma-1}\hat{W}_n - \phi\epsilon\hat{P}_s + \alpha\hat{P}_n - \alpha\hat{\tau}_{sn}. \tag{76}$$

When $\phi = 1$, there is no capital income, so that $s_n = 1$, and the intermediate-input share is $1-\xi$. Equation 76 therefore reduces to Equation 74. When $\phi = 0$, there is no intermediate-input use, and Equation 76 reduces to Equation 75.

M Two extreme cases

This appendix examines two limiting cases of the model to characterize how the equilibrium responses to a unilateral tariff change as key parameters approach their limiting values.

M.1 Case 1: σ approach to 1

When $\phi = 0$, i.e., when labor and mobile capital are the only production inputs, the log-linearized equilibrium conditions can be summarized by the following general matrix system.

Define

$$V_J \equiv \sum_I V_{JI} = V_{JJ} + V_{JL}, \qquad V_L \equiv \sum_I V_{LI} = V_{LJ} + V_{LL}.$$

The equilibrium system can then be written as

$$\begin{pmatrix} \xi\epsilon V_J & 0 & \frac{\alpha}{1-\sigma}V_{JJ} & \frac{\alpha}{1-\sigma}V_{JL} \\ 0 & \xi\epsilon V_L & \frac{\alpha}{1-\sigma}V_{LJ} & \frac{\alpha}{1-\sigma}V_{LL} \\ V_J & V_L & 0 & 0 \\ \xi V_J + (\xi\epsilon - 1)V_{JJ} & (\xi\epsilon - 1)V_{LJ} & \frac{\alpha}{1-\sigma}(V_{JJ} + V_{LJ}) & 0 \end{pmatrix} \begin{pmatrix} \hat{W}_J \\ \hat{W}_L \\ \hat{B}_J \\ \hat{B}_L \end{pmatrix} = \begin{pmatrix} -\alpha V_{JL}\hat{\tau}_{JL} \\ -\alpha V_{LJ}\hat{\tau}_{LJ} \\ 0 \\ -\alpha V_{LJ}\hat{\tau}_{LJ} \end{pmatrix}. \tag{M.1}$$

Recall that

$$\epsilon = \frac{\sigma\alpha}{\sigma - 1}.$$

Hence, as $\sigma \to 1^+$,

$$\epsilon \to \infty, \qquad \frac{\alpha}{1-\sigma} = -\frac{\epsilon}{\sigma}.$$

To characterize the limiting equilibrium system, divide the first, second, and fourth rows of Equation (M.1) by ϵ, while leaving the third row unchanged. This gives

$$\begin{pmatrix} \xi V_J & 0 & -\frac{1}{\sigma}V_{JJ} & -\frac{1}{\sigma}V_{JL} \\ 0 & \xi V_L & -\frac{1}{\sigma}V_{LJ} & -\frac{1}{\sigma}V_{LL} \\ V_J & V_L & 0 & 0 \\ \frac{\xi}{\epsilon}V_J + \left(\xi - \frac{1}{\epsilon}\right)V_{JJ} & \left(\xi - \frac{1}{\epsilon}\right)V_{LJ} & -\frac{1}{\sigma}(V_{JJ} + V_{LJ}) & 0 \end{pmatrix} \begin{pmatrix} \hat{W}_J \\ \hat{W}_L \\ \hat{B}_J \\ \hat{B}_L \end{pmatrix} = \begin{pmatrix} -\frac{\alpha}{\epsilon}V_{JL}\hat{\tau}_{JL} \\ -\frac{\alpha}{\epsilon}V_{LJ}\hat{\tau}_{LJ} \\ 0 \\ -\frac{\alpha}{\epsilon}V_{LJ}\hat{\tau}_{LJ} \end{pmatrix}. \tag{M.2}$$

Since

$$\frac{1}{\epsilon} \to 0, \qquad \frac{\alpha}{\epsilon} = \frac{\sigma - 1}{\sigma} \to 0, \qquad \frac{1}{\sigma} \to 1$$

as $\sigma \to 1^+$, the tariff disturbances on the right-hand side vanish. The limiting system is therefore

$$\underbrace{\begin{pmatrix} \xi V_J & 0 & -V_{JJ} & -V_{JL} \\ 0 & \xi V_L & -V_{LJ} & -V_{LL} \\ V_J & V_L & 0 & 0 \\ \xi V_{JJ} & \xi V_{LJ} & -(V_{JJ} + V_{LJ}) & 0 \end{pmatrix}}_{\equiv M} \begin{pmatrix} \hat{W}_J \\ \hat{W}_L \\ \hat{B}_J \\ \hat{B}_L \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}. \tag{M.3}$$

Using

$$V_J = V_{JJ} + V_{JL}, \qquad V_L = V_{LJ} + V_{LL},$$

the determinant of the limiting matrix M is

$$\begin{aligned}\det(M) = \xi\,&(V_{JJ} + V_{JL} + V_{LJ} + V_{LL}) \\ &\times \left[V_{JJ}V_{JL}V_{LJ} + V_{JJ}V_{JL}V_{LL} + V_{JJ}V_{LJ}V_{LL} + V_{JL}V_{LJ}V_{LL}\right]. \end{aligned} \tag{M.4}$$

For $\xi > 0$ and strictly positive bilateral sales,

$$V_{JJ}, V_{JL}, V_{LJ}, V_{LL} > 0,$$

it follows that

$$\det(M) > 0.$$

Hence, M is nonsingular. The homogeneous system in Equation (M.3) therefore has the unique solution

$$\hat{W}_J = \hat{W}_L = \hat{B}_J = \hat{B}_L = 0.$$

It follows that

$$\lim_{\sigma \to 1^+} \hat{W}_J = \lim_{\sigma \to 1^+} \hat{W}_L = 0. \tag{M.5}$$

Since the change in capital employed in each country is proportional to the change in domestic production, which in turn is proportional to the wage response when $\phi = 0$, the capital-flow response also vanishes:

$$\lim_{\sigma \to 1^+} \hat{K}_J = \lim_{\sigma \to 1^+} \hat{K}_L = 0. \tag{M.6}$$

To understand the intuition for why this effect vanishes, consider the log-differentiated value of Foreign sales to Home,

$$\hat{V}_{FH} = \hat{W}_F - \frac{\alpha}{\sigma - 1}(a\hat{W}_F - b\hat{W}_H) + \alpha\hat{P}_H - \alpha\hat{\tau}_{FH}.$$

Most directly, because the Home price index includes prices affected by the increase in trade costs but rises by less than the trade cost itself, the last two terms,

$$\alpha(\hat{P}_H - \hat{\tau}_{FH}) < 0,$$

put downward pressure on V_{FH}. The remaining terms capture the response of domestic and foreign wages. A higher trade cost makes Foreign firms less competitive in the Home market, putting downward pressure on the Foreign wage. The resulting decline in Foreign production costs lowers Foreign prices and, through the first term $\hat{W}_F$, mechanically puts downward pressure on the value of Foreign sales. At the same time, the Home wage increases as Home firms face less import competition. The decline in the Foreign wage and the increase in the Home wage therefore provide an offsetting force through the endogenous adjustment

in relative production costs, as captured by

$$-\frac{\alpha}{\sigma-1}(a\hat{W}_F - b\hat{W}_H) > 0.$$

The role of σ can be seen from the coefficient $\alpha/(\sigma-1)$. As σ approaches one, this coefficient becomes arbitrarily large, so an increasingly small change in relative wages is sufficient to generate a given offsetting effect on Foreign sales. In equilibrium, the relative-wage adjustment therefore shrinks toward zero, while its effect through the term above offsets the direct downward pressure from the trade-cost increase. Consequently, the response of Foreign sales, and ultimately the equilibrium wage and capital-flow responses, converges to zero.

M.2 Case 2: Small open economy

Suppose Home approaches the small-open-economy limit, so that a change in the Home tariff has a vanishing effect on Foreign equilibrium variables. Recall that the rental rate is chosen as the numeraire, so that

$$\hat{r} = 0$$

throughout. Consider the global capital-market-clearing condition,

$$rK_0 = (1-\xi)(V_H + V_F).$$

Since the global capital endowment K_0 is fixed and the rental rate is the numeraire, log differentiation gives

$$0 = \omega_H \hat{V}_H + (1-\omega_H)\hat{V}_F, \qquad \omega_H \equiv \frac{V_H}{V_H + V_F}.$$

In the small-open-economy limit,

$$\omega_H \to 0 \qquad \text{and} \qquad \hat{V}_F \to 0.$$

The global capital-market-clearing condition therefore places no restriction on the proportional response of Home production $\hat{V}_H$ in the limit. Although global capital-market clearing continues to hold, it no longer provides an equilibrium condition determining Home variables. Instead, Home takes Foreign equilibrium variables as given, and the Home equilibrium is determined by the remaining domestic equilibrium conditions.

To get $\hat{W}_H$ and $\hat{B}_H$, I solve the Home's labor market clearing condition:

$$V_H \hat{W}_H = V_{HF}\hat{V}_{HF} + V_{HH}\hat{V}_{HH}$$

and trade balance condition:

$$\xi V_H \hat{W}_H = V_{FH}\hat{V}_{FH} + V_{HH}\hat{V}_{HH}$$

where

$$\hat{V}_{HF} = (1 - \xi\epsilon)\hat{W}_H - \alpha\hat{\tau}_{HF}$$
$$\hat{V}_{HH} = (1 - \xi\epsilon)\hat{W}_H + \frac{\alpha}{\sigma - 1}\hat{B}_H$$
$$\hat{V}_{FH} = \frac{\alpha}{\sigma - 1}\hat{B}_H - \alpha\hat{\tau}_{FH}$$

Thus I have:

$$\begin{pmatrix} \xi\epsilon V_H & \frac{\alpha}{1-\sigma}V_{HH} \\ \xi V_H + (\xi\epsilon - 1)V_{HH} & \frac{\alpha}{1-\sigma}(V_{HH} + V_{FH}) \end{pmatrix} \begin{pmatrix} \hat{W}_H \\ \hat{B}_H \end{pmatrix} = \begin{pmatrix} -\alpha V_{HF}\hat{\tau}_{HF} \\ -\alpha V_{FH}\hat{\tau}_{FH} \end{pmatrix}. \tag{M.7}$$

A bilateral increase in trade costs gives

$$\begin{pmatrix} \hat{W}_H \\ \hat{B}_H \end{pmatrix} = \begin{pmatrix} -\alpha\hat{\tau}\dfrac{V_{HH}(V_{HF} - V_{FH}) + V_{FH}V_{HF}}{A} \\ (\sigma - 1)\hat{\tau}\dfrac{\epsilon\xi V_H V_{FH} - \xi V_H V_{HF} - \epsilon\xi V_{HF}V_{HH} + V_{HF}V_{HH}}{A} \end{pmatrix},$$

where

$$A \equiv \epsilon\xi V_H V_{FH} + \epsilon\xi V_H V_{HH} - \xi V_H V_{HH} - \epsilon\xi V_{HH}^2 + V_{HH}^2.$$

Under the usual parameter restrictions $0 < \xi < 1$ and $\epsilon \geq 1$, the denominator can be written as

$$A = (1 - \xi)V_{HH}^2 + \xi(\epsilon - 1)V_{HF}V_{HH} + \epsilon\xi V_H V_{FH} > 0.$$

For a bilateral increase in trade costs, $\hat{\tau} > 0$, the sign of the Home wage response is therefore determined by

$$\operatorname{sgn}(\hat{W}_H) = -\operatorname{sgn}\left[V_{HH}(V_{HF} - V_{FH}) + V_{FH}V_{HF}\right].$$

Thus,

$$\hat{W}_H < 0 \quad \Longleftrightarrow \quad V_{HF}(V_{HH} + V_{FH}) > V_{HH}V_{FH},$$

whereas the inequality is reversed when $\hat{W}_H > 0$.

Similarly,

$$\operatorname{sgn}(\hat{B}_H) = \operatorname{sgn}\left[\xi V_H(\epsilon V_{FH} - V_{HF}) + V_{HF}V_{HH}(1 - \epsilon\xi)\right].$$

Hence, in the small-open-economy case, neither response has an unconditional sign. Instead, the effects of a bilateral change in trade costs depend on the initial pattern of bilateral and domestic trade flows, as well as on the parameter combination $\epsilon\xi$.

N Exact hat equations

This section presents the full system of exact-hat equations used to characterize the equilibrium responses to changes in trade costs in the n-region quantitative model. The system

consists of the bilateral sales equations, firm-entry conditions, regional labor-market and trade-balance conditions, and the global capital-market clearing condition.

Suppose τ_{sn} is the policy variable. Using exact-hat algebra, where $\hat{X} \equiv X'/X$, sales from source region s to destination region n satisfy:

$$\hat{V}_{sn} = \hat{N}_s \hat{B}_n^{\frac{\alpha}{\sigma-1}} (\hat{W}_S^{\xi} \hat{P}_S^{(1-\xi)\phi})^{1-\epsilon} \widehat{1+\tau_{SI}}^{-\alpha} \tag{77}$$

where

$$\hat{N}_s = \hat{W}_s^{1-\xi} \hat{P}_s^{(1-\xi)\phi} \tag{78}$$

The market clearing conditions are thus, for each source region S, labor market clearing conditions are:

$$\sum_n V_{sn} \hat{W}_s = \sum_n V_{sn} \hat{V}_{sn} \tag{79}$$

For each destination region n, trade balane conditions are:

$$(\xi + \phi(1-\xi)) \sum_n V_{sn} \hat{W}_s + \gamma_s r K_0 = \sum_k V_{ks} \hat{V}_{ks} \tag{80}$$

and the one capital market clearing condition as:

$$r K_0 = (1-\xi)(1-\phi) \sum_s \sum_n V_{sn} \hat{W}_s \tag{81}$$